

CHAPTER 3

POSITION ANALYSIS OF PARALLEL MANIPULATORS

3.1 INTRODUCTION

In this chapter, the position analysis of a class of spatial multi-dof mechanisms known as *parallel manipulators* is studied. Parallel manipulators are classified as planar, spherical, or spatial manipulators in accordance with their motion characteristics. Then the direct and inverse kinematics of several parallel manipulators are analyzed.

As shown in Fig. 3.1, a parallel manipulator typically consists of a moving platform that is connected to a fixed base by several *limbs* or *legs*. Typically, the number of limbs is equal to the number of degrees of freedom such that every limb is controlled by one actuator and all the actuators can be mounted at or near the fixed base. For this reason, parallel manipulators are sometimes called *platform manipulators*. Because the external load can be shared by the actuators, parallel manipulators tend to have a large load-carrying capacity.

Parallel manipulators can be found in many applications, such as air-plane simulators (Stewart, 1965), adjustable articulated trusses (Reinholtz and Gokhale, 1987), mining machines (Arai et al., 1991), pointing devices (Gosselin and Hamel, 1994), and walking machines (Waldron et al., 1984). Recently, it has also been developed as a high-speed, high-precision, multi-dof machining center, shown in Fig. 3.1 (Giddings & Lewis, 1995).

The design of parallel manipulators can be dated back to 1962, when Gough and Whitehall (1962) devised a six-linear jack system for use as a universal tire-testing machine. Stewart (1965) designed a platform manipulator for use as an aircraft simulator in 1965. Hunt (1983) made a systematic

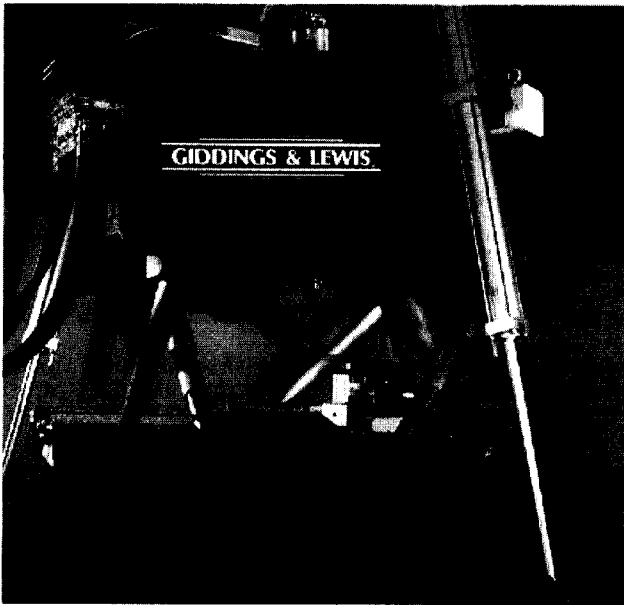


FIGURE 3.1. VARIAX[®] machining center. (Courtesy of Giddings & Lewis Machine Tools, Fond du Lac, Wisconsin.)

study of the kinematic structure of parallel manipulators. Since then, parallel manipulators have been studied extensively by numerous researchers (Clearly and Arai, 1991; Fichter, 1986; Grffis and Duffy, 1989; Innocenti and Parenti-Castelli, 1990; Mohamed and Duffy, 1985; Nanua et al., 1990; Zhang and Song, 1994).

Most of the 6-dof parallel manipulators studied to date consist of six extensible limbs. These parallel manipulators possess the advantages of high stiffness, low inertia, and large payload capacity. However, they suffer the problems of relatively small useful workspace and design difficulties. Furthermore, their direct kinematics is a very difficult problem. Perhaps, the only six-limbed, 6-dof parallel manipulators for which closed-form direct kinematic solutions have been reported in the literature are special forms of the Stewart–Gough platform (Chen and Song, 1994; Grffis and Duffy, 1989; Husain and Waldron, 1994; Lin et al., 1994; Nanua et al., 1990). In these special forms, pairs of concentric spherical joints may present manufacturing problems. As to the general Stewart–Gough platform, researchers have to resort to numerical techniques for the solutions. Raghavan (1993) applied the continuation method, while Innocenti and Parenti-Castelli (1993a) developed an exhaustive monodimensional search algorithm to solve the direct kinematics of the general Stewart–Gough platform.

Other variations of the Stewart–Gough platform have also been proposed in an attempt to overcome the aforementioned shortcomings. See Clavel (1988), Hudgens and Tesar (1988), Kohli et al. (1988), Lee and Shah (1987), Pierrot et al. (1990, 1991), Sternheim (1987), Tahmasebi and Tsai (1994, 1995), Tsai (1996), Tsai and Tahmasebi (1993), and Tsai et al. (1996) for examples.

3.2 STRUCTURE CLASSIFICATION OF PARALLEL MANIPULATORS

A parallel manipulator is said to be *symmetrical* if it satisfies the following conditions:

1. The number of limbs is equal to the number of degrees of freedom of the moving platform.
2. The type and number of joints in all the limbs are arranged in an identical pattern.
3. The number and location of actuated joints in all the limbs are the same.

When the conditions above are not satisfied, the manipulator is called *asymmetrical*. In this chapter we are concerned primarily with symmetrical manipulators. We observe that in a symmetrical manipulator the number of limbs, m , is equal to the number of degrees of freedom, F , which is also equal to the total number of loops, $L + 1$ (including the external loop); that is,

$$m = F = L + 1. \quad (3.1)$$

Let the *connectivity*, C_k , of a limb be the degrees of freedom associated with all the joints in the limb. It follows that

$$\sum_{k=1}^m C_k = \sum_{i=1}^j f_i, \quad (3.2)$$

where j is the number of joints in a mechanism. Substituting Eq. (1.8) into (3.2) and eliminating L by making use of (3.1), we obtain

$$\sum_{k=1}^m C_k = (\lambda + 1)F - \lambda. \quad (3.3)$$

Furthermore, the connectivity of each limb should not be greater than the motion parameter and less than the degrees of freedom of the moving plat-

form; that is,

$$\lambda \geq C_k \geq F. \quad (3.4)$$

Equations (3.3) and (3.4) are useful for enumeration and classification of parallel manipulators. Parallel mechanisms can generally be classified as planar, spherical, or spatial mechanisms. In what follows, we discuss the enumeration of each of these three types of manipulators.

3.2.1 Planar Parallel Manipulators

For planar 3-dof, three-limbed parallel manipulators, we have $\lambda = 3$ and $m = F = 3$. Substituting $\lambda = 3$ and $F = 3$ into Eq. (3.3), we obtain

$$C_1 + C_2 + C_3 = 4F - 3 = 9. \quad (3.5)$$

At the same time, Eq. (3.4) reduces to

$$3 \geq C_k \geq 3. \quad (3.6)$$

Hence the connectivity of each limb should be equal to 3; that is, each limb should have 3 degrees of freedom in its joints. Assuming that each limb consists of two links and three joints, each joint must be a 1-dof joint. Using revolute and prismatic joints as the kinematic pairs, we obtain seven possible limbs arrangements: RRR , RRP , RPR , PRR , RPP , PRP , and PPR . Further, if we limit ourselves to manipulators with three identical limb structures, only seven 3-dof planar parallel manipulators are feasible.

Figure 3.2 shows a planar 3-dof parallel manipulator using the $3RRR$ limb structure (Gosselin and Angeles, 1988); Figure 3.3 shows a planar 3-dof parallel manipulator using the $3PRP$ limb structure, where the three revolute joint axes are perpendicular to the plane of motion while the prismatic joint axes lie on the plane of motion (Mohammadi et al., 1993).

3.2.2 Spherical Parallel Manipulators

The motion parameter for spherical mechanisms is also equal to 3. Hence the connectivity requirement for spherical parallel manipulators is identical to that of planar parallel manipulators. For spherical linkage manipulators, the only permissible joint type is the revolute joint, and all the joint axes must intersect at a common point, called the *spherical center*. Hence the only possible limb structure is the RRR configuration. The manipulator shown in Fig. 1.21 is a 3-dof, $3RRR$ spherical parallel manipulator.

We note that one spherical joint can be installed at the center of a spherical parallel manipulator. However, such a spherical joint can only be used

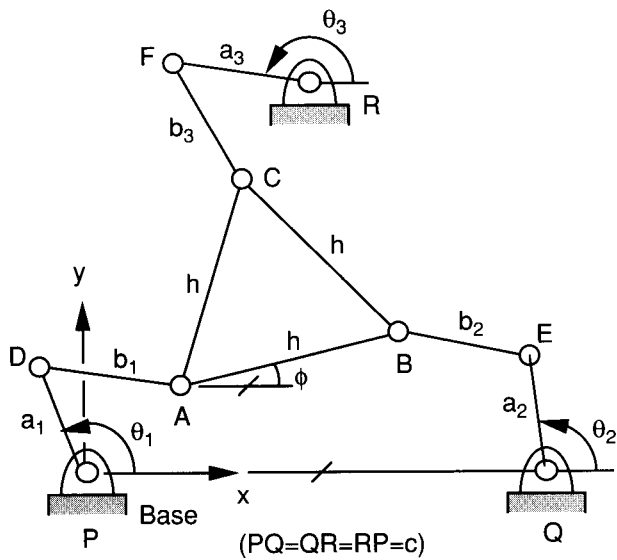


FIGURE 3.2. Planar 3-dof, 3RRR parallel manipulator.

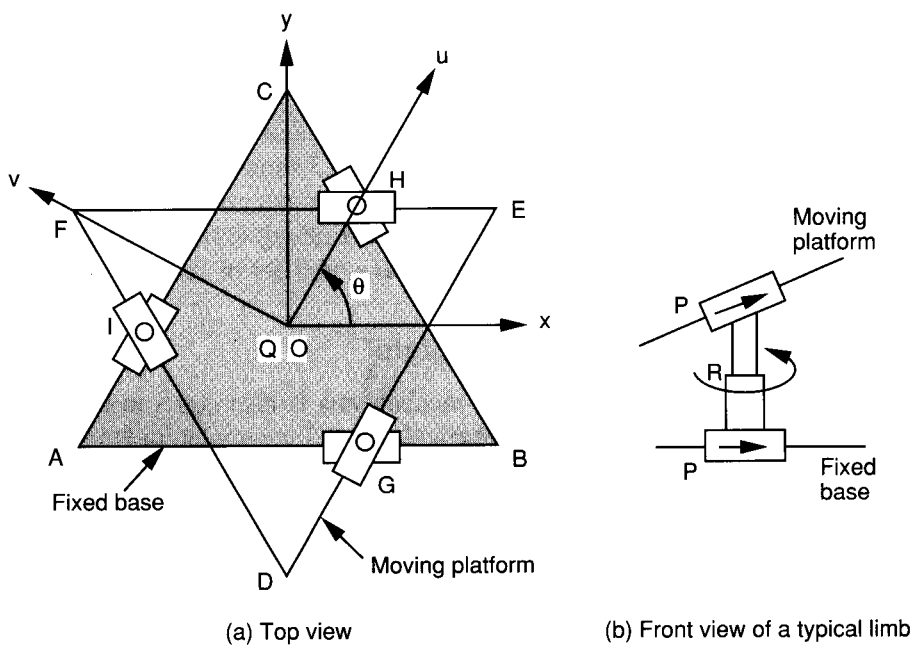


FIGURE 3.3. Planar 3-dof, 3PRP parallel manipulator.

as a passive joint, since it cannot be actuated by existing actuators. Hence, if a spherical joint is used, three additional limbs would be needed for parallel actuation of the moving platform. In this case, the number of limbs, including the limb with a spherical joint of zero length, is no longer equal to the number of degrees of freedom. Readers can refer to several articles related to the design and analysis of spherical parallel manipulators (Gosselin and Angeles, 1989; Gosselin and Hamel, 1994; Innocenti and Parenti-Castelli, 1993b; Wohlhart, 1994).

3.2.3 Spatial Parallel Manipulators

Substituting $\lambda = 6$ into Eqs. (3.3) and (3.4) for spatial mechanisms, we obtain

$$\sum_k^m C_k = 7F - 6 \quad (3.7)$$

and

$$6 \geq C_k \geq F. \quad (3.8)$$

Solving Eqs. (3.7) and (3.8) simultaneously for positive integers of C_k , $k = 1, 2, 3, \dots$, we can classify spatial parallel manipulators according to their degrees of freedom and connectivity listing as given in Table 3.1 (Hunt, 1983).

The number of links incorporated in each limb can be any as long as the sum of all joint freedoms is equal to the required connectivity. The maximum number of links occurs when all the joints are 1-dof joints. In practice, it is desirable to employ just two major links connecting the moving platform to

TABLE 3.1. Classification of Spatial Parallel Manipulators

Degrees of Freedom F	Number of Loops L	Sum of All Joint Freedoms $\sum_i f_i$	Connectivity Listing $C_k, k = 1, 2, 3, \dots$
2	1	8	4,4
			5,3
			6,2
3	2	15	5,5,5
			6,5,4
			6,6,3
4	3	22	6,6,5,5
			6,6,6,4
5	4	29	6,6,6,6,5
6	5	36	6,6,6,6,6,6

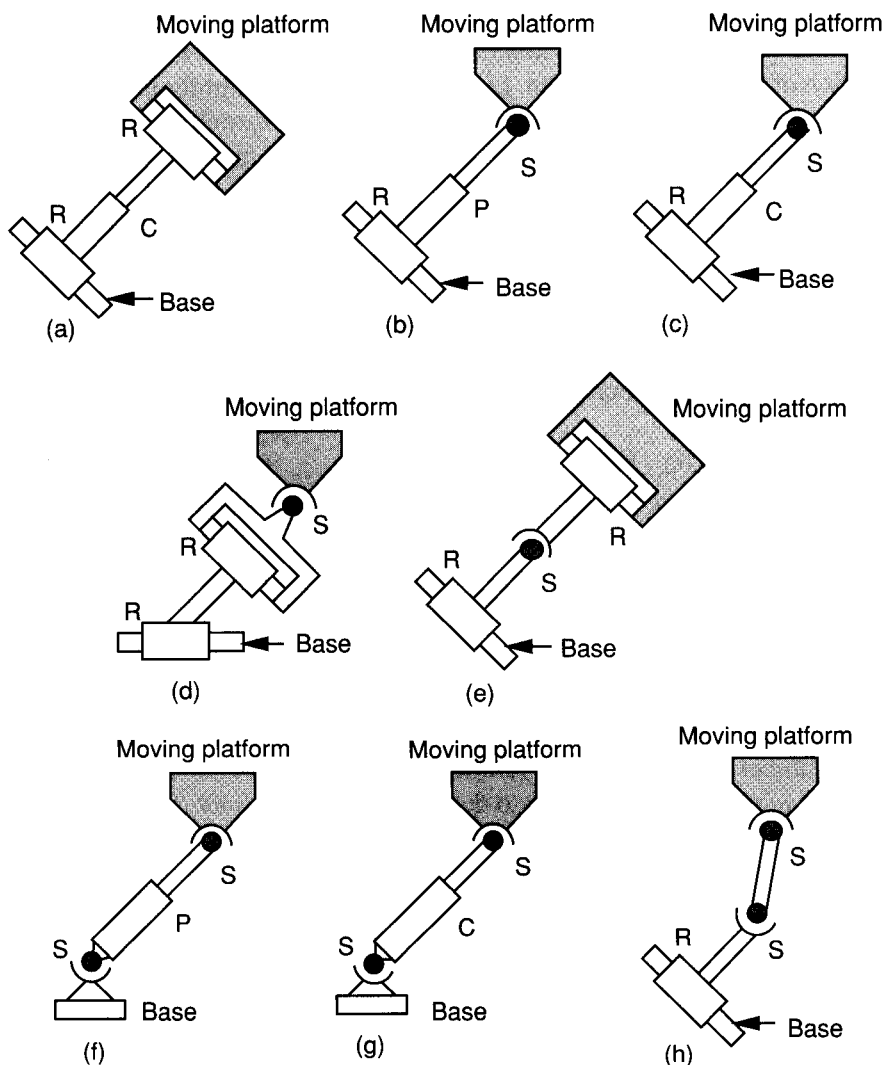


FIGURE 3.4. Eight possible limb configurations.

the base by three joints. Figure 3.4 illustrates a few such limb configurations. Figure 3.4a is a 4-dof limb, Fig. 3.4b–e are 5-dof limbs, and Fig. 3.4f–h are 6-dof limbs. Note that each of the limbs shown in Fig. 3.4c, f, and h contains 1 passive degree of freedom, while the one shown in Fig. 3.4g contains 2 passive degrees of freedom.

If limbs of identical kinematic structure are desired, the (4,4), (5,5,5), and (6,6,6,6,6,6) connectivity listings are the only feasible limb arrangements for

the 2-, 3-, and 6-dof parallel manipulators. For example, the 6-dof machining center developed by Giddings & Lewis (1995) shown in Fig. 3.1 has a connectivity listing of (6,6,6,6,6,6). Six *6UPS* limbs connect a moving platform to the fixed base. The prismatic joint in the middle of each limb is driven by a dc motor using a linear ball screw as the force transmission mechanism. Extending and retracting of the limbs alter the location of the moving platform on which a spindle is mounted, making six-axis machining feasible.

3.3 DENAVIT-HARTENBERG METHOD VERSUS GEOMETRIC METHOD

In this section we present the Denavit-Hartenberg matrix method of analysis. To facilitate the analysis, a Cartesian coordinate system is defined on each link in accordance with the Denavit-Hartenberg convention as shown in Fig. 3.5. Unlike open-loop manipulators, all the coordinate systems in a closed-loop mechanism are defined completely by the geometry of the linkage. In particular, the end-effector (n th) coordinate system coincides with the base (zeroth) coordinate system.

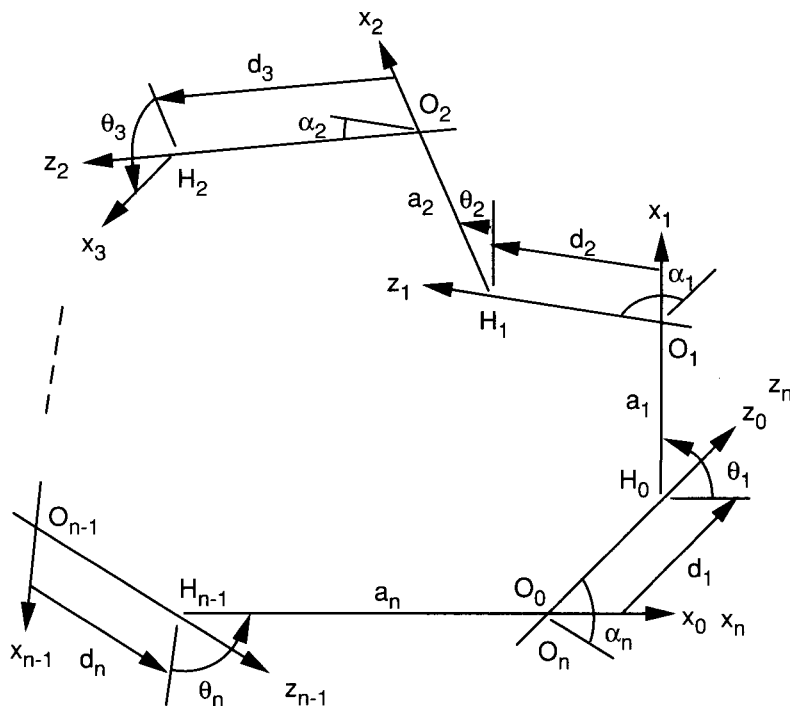


FIGURE 3.5. Closed-loop mechanism and its D-H parameters.

A change of coordinates between the i th coordinate system and the $(i - 1)$ th coordinate system is given in Eq. (2.2). When the change of coordinates is taken in succession, the corresponding matrices are multiplied. When all the changes of coordinates are taken, starting from the (x_0, y_0, z_0) system, going around the chain, and returning to the (x_0, y_0, z_0) system, the resulting matrix of transformation is given by ${}^0A_1 {}^1A_2 \cdots {}^{n-2}A_{n-1} {}^{n-1}A_n$. But this transformation is an identity transformation because of the return to the original system. Hence we can write

$${}^0A_1 {}^1A_2 \cdots {}^{n-2}A_{n-1} {}^{n-1}A_n = I, \quad (3.9)$$

where I is a 4×4 identity matrix.

For a multiple-loop mechanism, we formulate the loop-closure equation above once for each independent loop of the mechanism to obtain a set of constraint equations. We note that a universal joint can be modeled by two intersecting revolute joints, and a spherical joint can be modeled by three intersecting revolute joints.

Although the Denavit–Hartenberg method is very general, the method is complicated by the existence of multiple closed loops. For parallel manipulators, it is often more convenient to employ the geometric method. Generally, a *vector-loop equation* is written for each limb, and the passive joint variables are eliminated among these equations. In the following sections, we analyze the kinematics of several parallel manipulators to illustrate the methodology.

3.4 POSITION ANALYSIS OF A PLANAR 3RRR PARALLEL MANIPULATOR

For the planar 3-dof parallel manipulator shown in Fig. 3.2, the three fixed pivots P , Q , and R define the geometry of a fixed base, and the three moving pivots A , B , and C define the geometry of a moving platform. Three limbs connect the moving platform at points A , B , and C to the fixed base at points P , Q , and R by revolute joints. Each limb consists of two links connected by a revolute joint. Together, the mechanism consists of eight links and nine revolute joints. The degrees of freedom of the mechanism are computed from Eq. (1.4):

$$F = \lambda(n - j - 1) + \sum_i f_i = 3(8 - 9 - 1) + 9 = 3.$$

Hence the mechanism has 3 degrees of freedom. We consider the moving platform as the output link and links PD , QE , and RF as the input links. Thus all actuators can be installed on the fixed base.

3.4.1 Geometry of the Manipulator

Referring to Fig. 3.2, the origin of the fixed coordinate frame is located at point P . The x -axis points along the direction of $\overline{PQ}$ and the y -axis is perpendicular to $\overline{PQ}$. We assume that both the moving platform ABC and the fixed platform PQR are equilateral triangles (i.e., $AB = BC = AC = h$ and $PQ = QR = RP = c$). Figure 3.6 shows the link lengths and joint angles of a typical limb i .

The location of the moving platform can be specified in terms of the position of point A , and an orientation angle, ϕ . Then the coordinates of points B and C can be written in terms of A and ϕ as follows:

$$\begin{aligned} x_B &= x_A + hc\phi, \\ y_B &= y_A + hs\phi, \end{aligned} \quad (3.10)$$

and

$$\begin{aligned} x_C &= x_A + hc\left(\phi + \frac{\pi}{3}\right), \\ y_C &= y_A + hs\left(\phi + \frac{\pi}{3}\right). \end{aligned} \quad (3.11)$$

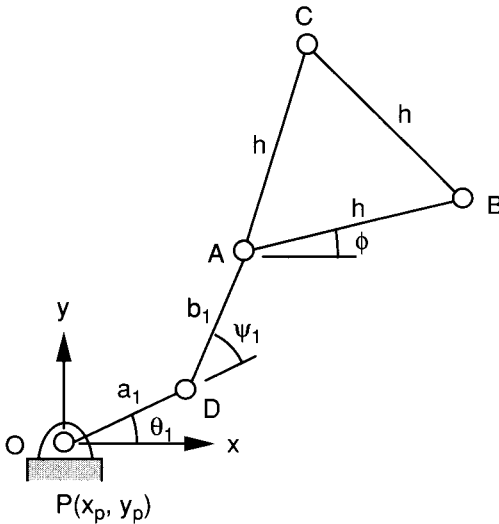


FIGURE 3.6. Joint angles associated with a typical limb of the 3RRR planar parallel manipulator.

From the geometry of Fig. 3.6, a vector-loop equation can be written as

$$\overline{OA} = \overline{OP} + \overline{PD} + \overline{DA}. \quad (3.12)$$

Expressing the vector-loop equation above in the fixed coordinate frame gives

$$\begin{aligned} x_A &= x_P + a_1 c\theta_1 + b_1 c(\theta_1 + \psi_1), \\ y_A &= y_P + a_1 s\theta_1 + b_1 s(\theta_1 + \psi_1). \end{aligned} \quad (3.13)$$

Since P is located at the origin, $x_P = y_P = 0$, identically. Since ψ_1 is a passive joint angle, it should be eliminated from the equations above. Toward this goal, we write Eq. (3.13) in the following form:

$$\begin{aligned} x_A - a_1 c\theta_1 &= b_1 c(\theta_1 + \psi_1), \\ y_A - a_1 s\theta_1 &= b_1 s(\theta_1 + \psi_1). \end{aligned} \quad (3.14)$$

Summing the squares of the two equations in (3.14) yields

$$x_A^2 + y_A^2 - 2x_A a_1 c\theta_1 - 2y_A a_1 s\theta_1 + a_1^2 - b_1^2 = 0. \quad (3.15)$$

Similarly, two additional equations can be derived for limbs 2 and 3:

$$\begin{aligned} x_A^2 + y_A^2 - 2x_A x_Q - 2y_A y_Q + x_Q^2 + y_Q^2 + h^2 + a_2^2 - b_2^2 + 2x_A h c\phi \\ + 2y_A h s\phi - 2x_A a_2 c\theta_2 - 2y_A a_2 s\theta_2 - 2a_2 h c\phi c\theta_2 - 2x_Q h c\phi - 2y_Q h s\phi \\ + 2x_Q a_2 c\theta_2 + 2y_Q a_2 s\theta_2 - 2a_2 h s\phi s\theta_2 = 0, \end{aligned} \quad (3.16)$$

$$\begin{aligned} x_A^2 + y_A^2 - 2x_A x_R - 2y_A y_R + x_R^2 + y_R^2 + h^2 + a_3^2 - b_3^2 \\ + 2x_A h c\left(\phi + \frac{\pi}{3}\right) + 2y_A h s\left(\phi + \frac{\pi}{3}\right) - 2x_A a_3 c\theta_3 \\ - 2y_A a_3 s\theta_3 - 2a_3 h c\left(\phi + \frac{\pi}{3}\right) c\theta_3 \\ - 2x_R h c\left(\phi + \frac{\pi}{3}\right) - 2y_R h s\left(\phi + \frac{\pi}{3}\right) + 2x_R a_3 c\theta_3 \\ + 2y_R a_3 s\theta_3 - 2a_3 h s\left(\phi + \frac{\pi}{3}\right) s\theta_3 = 0. \end{aligned} \quad (3.17)$$

3.4.2 Inverse Kinematics

For the inverse kinematics, x_A , y_A , and ϕ are given, and the joint angles θ_1 , θ_2 , and θ_3 are to be found. This can be accomplished on a limb-by-limb basis. For

limb 1 we arrange Eq. (3.15) in the following form:

$$e_1 s\theta_1 + e_2 c\theta_1 + e_3 = 0, \quad (3.18)$$

where

$$\begin{aligned} e_1 &= -2y_A a_1, \\ e_2 &= -2x_A a_1, \\ e_3 &= x_A^2 + y_A^2 + a_1^2 - b_1^2. \end{aligned}$$

Substituting the trigonometric identities

$$s\theta_i = \frac{2t_i}{1+t_i^2} \quad \text{and} \quad c\theta_i = \frac{1-t_i^2}{1+t_i^2}, \quad \text{where} \quad t_i = \tan \frac{\theta_i}{2}$$

into Eq. (3.18), we obtain

$$(e_3 - e_2)t_1^2 + 2e_1 t_1 + (e_3 + e_2) = 0. \quad (3.19)$$

Solving Eq. (3.19) for t_1 yields

$$\theta_1 = 2 \tan^{-1} \frac{-e_1 \pm \sqrt{e_1^2 + e_2^2 - e_3^2}}{e_3 - e_2}. \quad (3.20)$$

Hence, corresponding to each given moving platform location, there are generally two solutions of θ_1 and therefore two configurations of limb 1. When Eq. (3.18) yields a double root, the two links PD and DA are in a fully stretched-out or folded-back configuration called the *singular configuration*. When Eq. (3.18) yields no real root, the specified moving platform location is not reachable. Once θ_1 is solved, ψ_1 can be found from Eq. (3.13) by back substitution.

Following the same procedure, the other two limb configurations can be solved. We conclude that, in general, there are a total of eight possible manipulator postures corresponding to a given end-effector location.

3.4.3 Direct Kinematics

For the direct kinematics, θ_1 , θ_2 , and θ_3 are given, and the position (x_A, y_A) and the orientation angle ϕ of the moving platform are to be found. This can be accomplished by eliminating x_A and y_A from Eqs. (3.15), (3.16), and (3.17) as follows. First, we write Eqs. (3.15), (3.16), and (3.17) in the follow-

ing forms:

$$x_A^2 + y_A^2 + e_{11}x_A + e_{12}y_A + e_{13} = 0, \quad (3.21)$$

$$x_A^2 + y_A^2 + e_{21}x_A + e_{22}y_A + e_{23} = 0, \quad (3.22)$$

$$x_A^2 + y_A^2 + e_{31}x_A + e_{32}y_A + e_{33} = 0, \quad (3.23)$$

where

$$e_{11} = -2a_1c\theta_1,$$

$$e_{12} = -2a_1s\theta_1,$$

$$e_{13} = a_1^2 - b_1^2,$$

$$e_{21} = -2x_Q + 2hc\phi - 2a_2c\theta_2,$$

$$e_{22} = -2y_Q + 2hs\phi - 2a_2s\theta_2,$$

$$e_{23} = x_Q^2 + y_Q^2 + h^2 + a_2^2 - b_2^2 - 2a_2hc\phi c\theta_2 - 2a_2hs\phi s\theta_2 \\ - 2x_Qhc\phi - 2y_Qhs\phi + 2x_Qa_2c\theta_2 + 2y_Qa_2s\theta_2,$$

$$e_{31} = -2x_R + 2hc\left(\phi + \frac{\pi}{3}\right) - 2a_3c\theta_3,$$

$$e_{32} = -2y_R + 2hs\left(\phi + \frac{\pi}{3}\right) - 2a_3s\theta_3,$$

$$e_{33} = x_R^2 + y_R^2 + h^2 + a_3^2 - b_3^2 - 2a_3hc\left(\phi + \frac{\pi}{3}\right)c\theta_3 - 2a_3hs\left(\phi + \frac{\pi}{3}\right)s\theta_3 \\ - 2x_Rhc\left(\phi + \frac{\pi}{3}\right) - 2y_Rhs\left(\phi + \frac{\pi}{3}\right) + 2x_Ra_3c\theta_3 + 2y_Ra_3s\theta_3.$$

Note that e_{11} , e_{12} , and e_{13} are constants, while e_{21} , e_{22} , e_{23} , e_{31} , e_{32} , and e_{33} are linear functions of $s\phi$ and $c\phi$.

Equations (3.21), (3.22), and (3.23) constitute three nonlinear equations in three unknowns, x_A , y_A , and ϕ . This system of equations can be simplified by performing the following operations. Subtracting Eq. (3.22) from (3.21) yields

$$e'_{11}x_A + e'_{12}y_A + e'_{13} = 0. \quad (3.24)$$

Subtracting Eq. (3.23) from (3.21) yields

$$e'_{21}x_A + e'_{22}y_A + e'_{23} = 0. \quad (3.25)$$

Here $e'_{11} = e_{11} - e_{21}$, $e'_{12} = e_{12} - e_{22}$, $e'_{13} = e_{13} - e_{23}$, $e'_{21} = e_{11} - e_{31}$, $e'_{22} = e_{12} - e_{32}$, and $e'_{23} = e_{13} - e_{33}$ are linear functions of $\sin\phi$ and $\cos\phi$.

Equations (3.21) together with (3.24) and (3.25) form a new system of equations. We may solve Eqs. (3.24) and (3.25) for x_A and y_A and then sub-

stitute the resulting expressions into Eq. (3.21). This results in a fourth-degree polynomial in $s\phi$ and $c\phi$:

$$\delta_1^2 + \delta_2^2 + e_{11}\delta\delta_1 + e_{12}\delta\delta_2 + e_{13}\delta^2 = 0, \quad (3.26)$$

where

$$\begin{aligned} \delta &= e'_{11}e'_{22} - e'_{12}e'_{21}, \\ \delta_1 &= e'_{12}e'_{23} - e'_{13}e'_{22}, \\ \delta_2 &= e'_{13}e'_{21} - e'_{11}e'_{23}. \end{aligned}$$

Equation (3.26) can be converted into an eighth-degree polynomial by using the half-tangent angle expressions. Hence, corresponding to each given set of input joint angles, there are at most eight possible manipulator configurations. A numerical method of solution can be found in Gosselin and Sefrioui (1991).

3.5 POSITION ANALYSIS OF A SPATIAL ORIENTATION MECHANISM

In this section the kinematics of a 3-dof orientation platform is studied. First, we describe the kinematic structure of the manipulator. Then the inverse and direct kinematics are solved in closed form.

Figure 3.7 shows a spatial parallel manipulator that is made up of a moving platform, a fixed base, and three extensible limbs. Both the moving platform $OB_1B_2B_3$ and the fixed base $OA_1A_2A_3$ take the form of a tetrahedron. The moving platform is connected directly to the fixed base by a spherical joint at point O . In addition, three extensible limbs connect the moving platform at points B_i to the fixed base at points A_i by spherical joints. Each limb consists of an upper member and a lower member connected by a prismatic joint. Ball screws or hydraulic jacks can be used to vary the lengths of the prismatic joints and therefore to control the motion of the moving platform. Overall, there are eight links connected together by seven spherical and three prismatic joints. We note that this mechanism is not a spherical mechanism, because the three limbs and the moving platform do not have a common stationary point. In fact, the upper members of the three limbs do not possess spherical motions at all.

Substituting $\lambda = 6$, $n = 8$, $j = 10$, and the appropriate f_i for each joint into Eq. (1.4) yields

$$F = \lambda(n - j - 1) + \sum_i f_i = 6(8 - 10 - 1) + (7 \times 3 + 3) = 6.$$

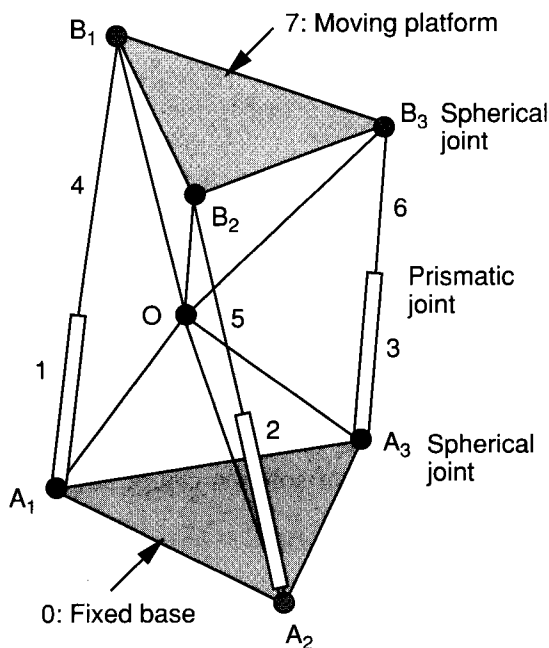


FIGURE 3.7. A 3-dof orientation platform.

However, there is a passive degree of freedom associated with each of the three limbs. Hence the moving platform possesses only 3 degrees of freedom. Although the motion of the whole mechanism is not spherical, the moving platform does possess a spherical motion because of the existence of a fixed point O . This mechanism can be used as an orienting device in which the moving platform is treated as the output link, and the three prismatic joints are employed as the input means.

3.5.1 Geometry of the Manipulator

For the purpose of analysis, two Cartesian coordinate systems $A(x, y, z)$ and $B(u, v, w)$ are attached to the fixed base and moving platform, respectively, as shown in Fig. 3.8. Without losing generality, we assume that the origin of frame A is located at the fixed point O , the z -axis points along the direction of $\overline{OA_1}$, the x -axis lies on the plane of OA_1A_2 , and the y -axis is defined according to the right-hand rule. Similarly, we assume that the origin of frame B is located at the fixed point O , the w -axis points along the direction of $\overline{OB_1}$, the u -axis lies on the plane OB_1B_2 , and the v -axis is defined according to the right-hand rule.

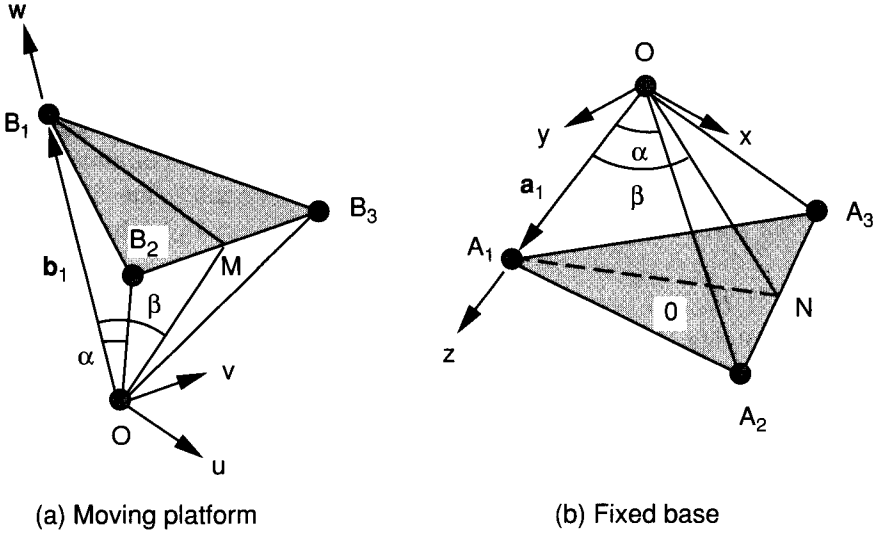


FIGURE 3.8. Coordinate systems of the orientation mechanism.

The transformation from the moving frame B to the fixed frame A can be described by a 3×3 rotation matrix ${}^A R_B$ defined by the following three Euler angles. Assume that the initial location of the moving frame B coincides with the fixed frame A and that the final location is obtained by a rotation of ϕ about the w -axis, followed by a second rotation of θ about the displaced u -axis, followed by a third rotation of ψ about the displaced w -axis. Then the resulting rotation matrix is given by:

$${}^A R_B = \begin{bmatrix} c\phi c\psi - s\phi c\theta s\psi & -c\phi s\psi - s\phi c\theta c\psi & s\phi s\theta \\ s\phi c\psi + c\phi c\theta s\psi & -s\phi s\psi + c\phi c\theta c\psi & -c\phi s\theta \\ s\theta s\psi & s\theta c\psi & c\theta \end{bmatrix}. \quad (3.27)$$

Let the position vector of point B_i with respect to the moving frame B be given by

$${}^B \mathbf{b}_i = [b_{iu}, b_{iv}, b_{iw}]^T. \quad (3.28)$$

Then the position vector of point B_i expressed in the fixed frame A is

$$\begin{aligned} \mathbf{b}_i &= [b_{ix}, b_{iy}, b_{iz}]^T = {}^A R_B {}^B \mathbf{b}_i \\ &= \begin{bmatrix} b_{iu}(c\phi c\psi - s\phi c\theta s\psi) - b_{iv}(c\phi s\psi + s\phi c\theta c\psi) + b_{iw}s\phi s\theta \\ b_{iu}(s\phi c\psi + c\phi c\theta s\psi) - b_{iv}(s\phi s\psi - c\phi c\theta c\psi) - b_{iw}c\phi s\theta \\ b_{iu}s\theta s\psi + b_{iv}s\theta c\psi + b_{iw}c\theta \end{bmatrix}. \end{aligned} \quad (3.29)$$

Also let the position vector of point A_i with respect to the fixed frame A be given by

$$\mathbf{a}_i = [a_{ix}, a_{iy}, a_{iz}]^T. \quad (3.30)$$

Then a loop-closure equation can be written for limb i as

$$\mathbf{d}_i = \mathbf{b}_i - \mathbf{a}_i, \quad (3.31)$$

where $\mathbf{d}_i = \overline{A_i B_i}$.

Dot-multiplying Eq. (3.31) with itself produces an equation of constraint imposed by limb i as follows:

$$d_i^2 = \mathbf{a}_i^2 + \mathbf{b}_i^2 - 2\mathbf{a}_i^T \mathbf{b}_i, \quad (3.32)$$

where d_i denotes the length of the i th limb. Substituting Eqs. (3.29) and (3.30) into (3.32) yields

$$e_{i1}c\phi + e_{i2}s\phi + e_{i3} = 0 \quad \text{for } i = 1, 2, 3, \quad (3.33)$$

where

$$e_{i1} = \lambda_{i1}c\psi + \lambda_{i2}s\psi + \lambda_{i3}, \quad (3.34)$$

$$e_{i2} = \mu_{i1}c\psi + \mu_{i2}s\psi + \mu_{i3}, \quad (3.35)$$

$$e_{i3} = v_{i1}c\psi + v_{i2}s\psi + v_{i3}, \quad (3.36)$$

$$\lambda_{i1} = a_{iy}b_{iv}c\theta + a_{ix}b_{iu},$$

$$\lambda_{i2} = a_{iy}b_{iu}c\theta - a_{ix}b_{iv},$$

$$\lambda_{i3} = -a_{iy}b_{iw}s\theta,$$

$$\mu_{i1} = -a_{ix}b_{iv}c\theta + a_{iy}b_{iu},$$

$$\mu_{i2} = -a_{ix}b_{iu}c\theta - a_{iy}b_{iv},$$

$$\mu_{i3} = a_{ix}b_{iw}s\theta,$$

$$v_{i1} = a_{iz}b_{iv}s\theta,$$

$$v_{i2} = a_{iz}b_{iu}s\theta,$$

$$v_{i3} = a_{iz}b_{iw}c\theta - \frac{1}{2}(\mathbf{a}_i^2 + \mathbf{b}_i^2 - d_i^2).$$

Equation (3.32) or (3.33) written three times, once for each $i = 1, 2$, and 3 , yields three equations of constraint for the moving platform.

3.5.2 Inverse Kinematics

For inverse kinematics, the orientation of the moving platform is known, and the problem is to find the limb lengths d_1 , d_2 , and d_3 . The orientation of the moving platform is described by the three Euler's angles, ϕ , θ , and ψ . Once the orientation of the moving platform is known, the position of points B_i for $i = 1, 2$, and 3 can be computed from Eq. (3.29). Hence the corresponding limb lengths d_i , $i = 1, 2, 3$, can be computed from Eq. (3.32):

$$d_i^2 = \pm \sqrt{\mathbf{a}_i^2 + \mathbf{b}_i^2 - 2\mathbf{a}_i^T \mathbf{b}_i} \quad \text{for } i = 1, 2, 3. \quad (3.37)$$

There are generally two solutions of d_i . However, only the positive limb length is physically realizable.

3.5.3 Direct Kinematics

For direct kinematics, the limb lengths d_1 , d_2 , and d_3 are given, and the problem is to find the orientation of the moving platform. Equation (3.33) written three times yields a system of three trigonometric equations that can be solved for the three variables ϕ , θ , and ψ . For the coordinate systems chosen, we have $a_{1x} = a_{1y} = a_{2y} = 0$, $b_{1u} = b_{1v} = b_{2v} = 0$, $a_{1z} = \ell_1 = OA_1$, and $b_{1w} = \ell_2 = OB_1$. Equation (3.33) written for $i = 1$ yields

$$\theta = \cos^{-1} \frac{\ell_1^2 + \ell_2^2 - d_2^2}{2\ell_1\ell_2}. \quad (3.38)$$

Hence Eq. (3.38) yields two solutions of θ . However, these two solutions represent only one manipulator orientation. Hence they will be counted as one solution. This is a common problem associated with the Euler angle representation of the orientation of a rigid body.

Once θ is found, we can write Eq. (3.33) twice for $i = 2$ and 3 and solve them for $c\phi$ and $s\phi$ as follows:

$$c\phi = \frac{e_{22}e_{33} - e_{23}e_{32}}{e_{21}e_{32} - e_{22}e_{31}}, \quad (3.39)$$

$$s\phi = \frac{e_{23}e_{31} - e_{21}e_{33}}{e_{21}e_{32} - e_{22}e_{31}}, \quad (3.40)$$

provided that $e_{21}e_{32} - e_{22}e_{31} \neq 0$. Then we eliminate ϕ by substituting Eqs. (3.39) and (3.40) into the trigonometric identity $c^2\phi + s^2\phi = 1$. This results in the following equation:

$$(e_{22}e_{33} - e_{23}e_{32})^2 + (e_{23}e_{31} - e_{21}e_{33})^2 - (e_{21}e_{32} - e_{22}e_{31})^2 = 0. \quad (3.41)$$

Substituting Eqs. (3.34)–(3.36) into Eq. (3.41) and making use of the trigonometric identities

$$c\psi = \frac{1-t^2}{1+t^2} \quad \text{and} \quad s\psi = \frac{2t}{1+t^2}, \quad \text{where} \quad t = \tan \frac{\psi}{2},$$

we obtain an eighth-degree polynomial in t :

$$\kappa_0 t^8 + \kappa_1 t^7 + \cdots + \kappa_7 t + \kappa_8 = 0, \quad (3.42)$$

where κ_i , $i = 0, 1, \dots, 8$, are functions of the geometric parameters $\mathbf{a}_i$ and $\mathbf{b}_i$, limb lengths d_i , and the angle θ . It follows that corresponding to each solution of θ , there are at most eight solutions of ψ . Once θ and ψ are known, Eqs. (3.39) and (3.40) yield a single value of ϕ . We conclude that the direct kinematics of the orientation platform has at most eight feasible postures.

3.6 POSITION ANALYSIS OF THE UNIVERSITY OF MARYLAND MANIPULATOR

In this section the kinematics of a 3-dof parallel manipulator developed at the University of Maryland (Tsai and Stamper, 1996) is studied. This manipulator, shown in Fig. 3.9, employs only revolute joints to constrain the moving platform output to translational motion. A schematic diagram of the manipulator is sketched in Fig. 3.10, where the fixed base is labeled as link 0 and the moving platform is labeled as link 16. Three identical limbs connect the moving platform to the fixed base. Each limb consists of an upper arm and a lower arm. The lower arms are labeled as links 1, 2, and 3. Each upper arm is made up of a planar four-bar parallelogram: links 4, 7, 10, and 13 for the first limb; 5, 8, 11, and 14 for the second limb; and 6, 9, 12, and 15 for the third limb. For each limb, the upper and lower arms and the two platforms are connected by three parallel revolute joints at points A , B , and C . The ground-connected joint axes lie on a plane. Similarly, the moving platform connected joint axes also lie on a plane. Furthermore, the two joint axes at B and C intersect the axes of the four-bar parallelogram perpendicularly. A more general configuration includes a small offset in each of links 4, 5, 6, 13, 14, and 15 (Tsai and Stamper, 1996). Revolute joints are used for the parallelogram as opposed to ball joints employed in the DELTA robot (Pierrot et al., 1990). Overall, there are 17 links and 21 revolute joints. The general degrees-of-freedom equation predicts that

$$F = \lambda(n - j - 1) + \sum_i f_i = 6(17 - 21 - 1) + 21 = -9.$$

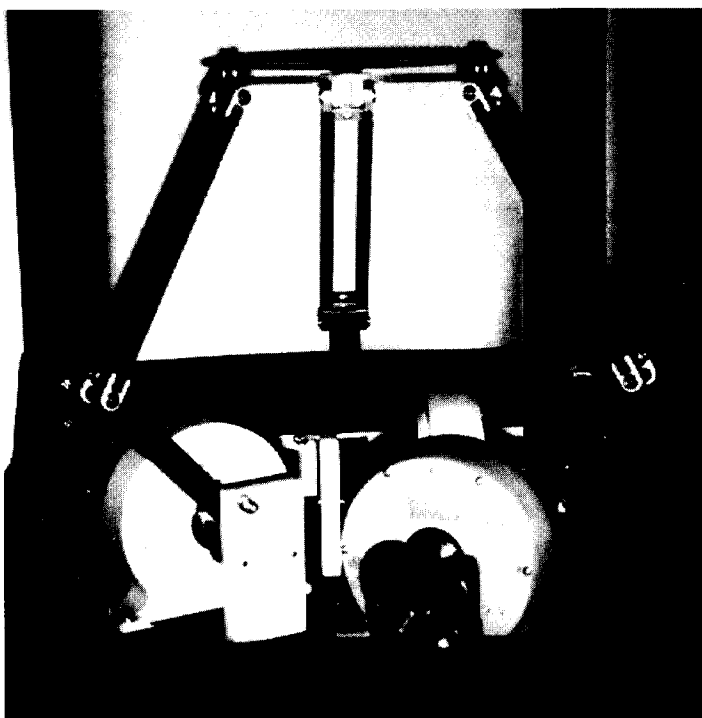


FIGURE 3.9. University of Maryland manipulator.

However, due to the special arrangement of the links and joints, many of the constraints imposed by the joints are redundant, and the resulting mechanism does have 3 translational degrees of freedom. Due to the four-bar parallelogram and the three parallel revolute joints at points A , B , and C , any single limb constrains the moving platform from rotating about the z and x_i axes. Hence any two limbs constrain the moving platform from rotating about any axis. This leaves the mechanism with 3 translational degrees of freedom and forces the moving platform to remain in the same orientation at all times. This unique characteristic is useful in many applications, such as an x - y - z positioning device. A hybrid manipulator can also be constructed by mounting a wrist mechanism onto the moving platform.

3.6.1 Geometry of the Manipulator

As shown in Fig. 3.10, a reference coordinate system (x, y, z) is attached to the center O of the fixed platform, with its x and y axes lie on the fixed plane and the z -axis points up vertically. Another coordinate system (x_i, y_i, z_i) is attached to the fixed base at point A_i , such that the x_i -axis is in line with the

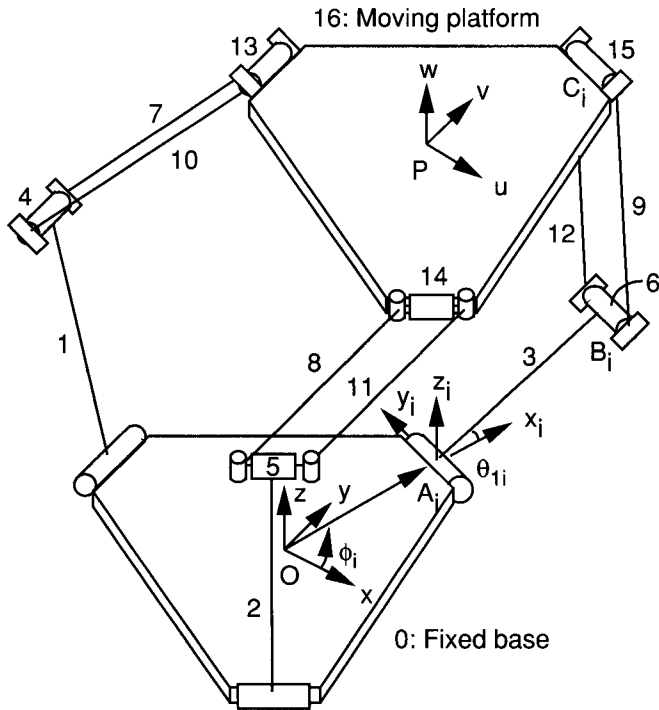


FIGURE 3.10. Platform manipulator with only translational degrees of freedom.

extended line of $\overline{OA}$, the y_i -axis is directed along the revolute joint axis at A_i , and the z_i -axis is parallel to the z -axis. The angle ϕ_i is measured from the x -axis to the x_i -axis and is a constant parameter of the manipulator design. Figure 3.11 defines the joint angles associated with the i th limb, wherein $\mathbf{p}$ is the position vector of the centroid of the moving platform, θ_{1i} is measured from the x_i axis to $\overline{AB}$, θ_{2i} is defined from the extended line of $\overline{AB}$ to the line defined by the intersection of the plane of the parallelogram and the x_i – z_i plane, and θ_{3i} is measured from the y_i direction to $\overline{B_iC_i}$. Overall, there are nine joint angles, θ_{1i} , θ_{2i} , and θ_{3i} for $i = 1, 2$, and 3 , associated with the manipulator.

For this manipulator, θ_{11} , θ_{12} , and θ_{13} are considered as the actuated joints. Other combinations of actuated joints are also possible, but actuating θ_{11} , θ_{12} , and θ_{13} offers the advantage of attaching all actuators to the fixed base. A loop-closure equation can be written for each limb:

$$\overline{A_iB_i} + \overline{B_iC_i} = \overline{OP} + \overline{PC_i} - \overline{OA_i}. \quad (3.43)$$

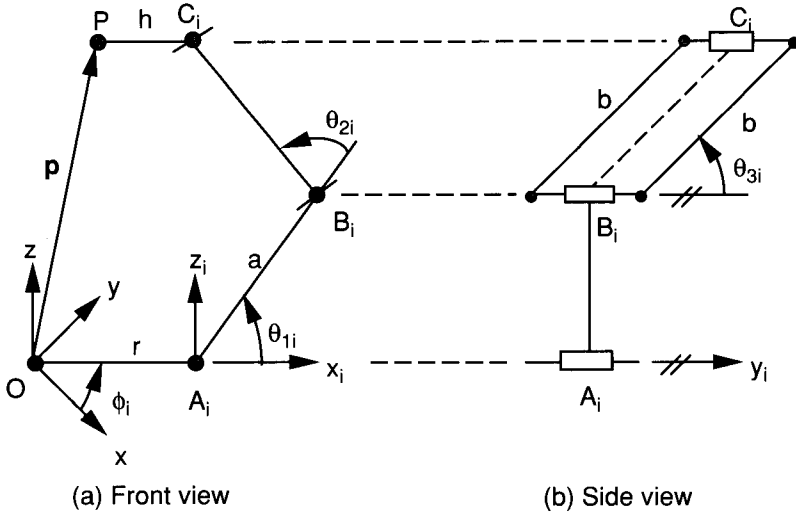


FIGURE 3.11. Description of the joint angles.

Expressing Eq. (3.43) in the (x_i, y_i, z_i) coordinate frame, we obtain

$$\begin{bmatrix} ac\theta_{1i} + bs\theta_{3i}c(\theta_{1i} + \theta_{2i}) \\ bc\theta_{3i} \\ as\theta_{1i} + bs\theta_{3i}s(\theta_{1i} + \theta_{2i}) \end{bmatrix} = \begin{bmatrix} c_{xi} \\ c_{yi} \\ c_{zi} \end{bmatrix}, \quad (3.44)$$

where

$$\begin{bmatrix} c_{xi} \\ c_{yi} \\ c_{zi} \end{bmatrix} = \begin{bmatrix} c\phi_i & s\phi_i & 0 \\ -s\phi_i & c\phi_i & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} + \begin{bmatrix} h - r \\ 0 \\ 0 \end{bmatrix} \quad (3.45)$$

denotes the position of point C_i relative to the (x_i, y_i, z_i) coordinate frame, a and b are the lengths of links $A_i B_i$ and $B_i C_i$, respectively, and $\mathbf{p} = [p_x, p_y, p_z]^T$ is the position vector of point P relative to the (x, y, z) coordinate system.

3.6.2 Inverse Kinematics

For inverse kinematics, the position vector $\mathbf{p}$ of the moving platform is given and the problem is to find the joint angles θ_{11} , θ_{12} , and θ_{13} required to bring the moving platform to the desired position. An intuitive approach for the solution is to consider the problem geometrically. We notice that once the position of P is given, the position of C_i is also known. Now consider the surface

generated by the full range of motion of $C_i B_i$ about point C . It is a sphere centered at point C_i . The full range of motion of $A_i B_i$ is a circle centered about A_i in the plane of motion of $A_i B_i$. The solution of the inverse kinematics problem is found at the intersection of this circle and the sphere as shown in Fig. 3.12. Four cases are possible:

1. *Generic solution.* The circle penetrates the sphere, resulting in two solutions.
2. *Singular solution.* The circle is tangent to the sphere, resulting in one solution.
3. *Singular solution.* The circle lies on the sphere, producing an infinite number of solutions. This is unlikely since it requires the moving and stationary platforms to occupy the same plane simultaneously.
4. *No real solution.* The circle and the sphere may not intersect at all.

With the geometry of the problem in mind, algebraic solutions are developed by solving Eq. (3.44). Two solutions of θ_{3i} are found by solving the second element of Eq. (3.44):

$$\theta_{3i} = \cos^{-1} \frac{c_{yi}}{b}. \quad (3.46)$$

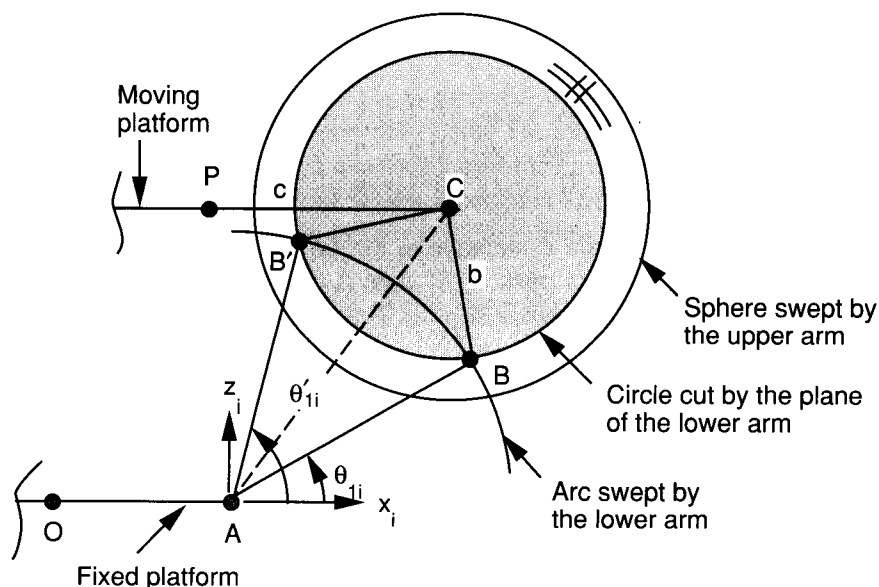


FIGURE 3.12. Two inverse kinematic solutions of a limb.

However, it will be shown later that either of the two solutions for θ_{3i} results in the same physical pose for limb i .

With θ_{3i} determined, an equation with θ_{2i} as the only unknown is generated by summing the squares of c_{xi} , c_{yi} , and c_{zi} in Eq. (3.44):

$$2abs\theta_{3i}c\theta_{2i} + a^2 + b^2 = c_{xi}^2 + c_{yi}^2 + c_{zi}^2. \quad (3.47)$$

Hence

$$\theta_{2i} = \cos^{-1} \kappa. \quad (3.48)$$

where $\kappa = (c_{xi}^2 + c_{yi}^2 + c_{zi}^2 - a^2 - b^2)/(2abs\theta_{3i})$. Therefore, corresponding to each solution of θ_{3i} , Eq. (3.48) yields two solutions of θ_{2i} . This results in four solution sets for θ_{2i} and θ_{3i} .

Furthermore, corresponding to each solution set of θ_{2i} and θ_{3i} , Eq. (3.44) yields a unique solution for θ_{1i} . However, the four solution sets contain only two distinct values of θ_{1i} . Moreover, for each solution of θ_{1i} , the two associated values of θ_{2i} differ by π , while the sum of the two associated values of θ_{3i} is 0. By virtue of these relationships and the geometry of the limb, each limb assumes the same physical pose for each θ_{1i} . Hence for each limb the four solution sets are realized in only two distinct poses.

The inverse kinematics solution is tested for special cases by examining Eq. (3.48). If $|c_{yi}| < b$ and $|\kappa| < 1$, the two solutions correspond to the configuration where the circle swept by $A_i B_i$ intersects the sphere swept by $C_i B_i$ in two locations. If $|c_{yi}| = b$ and $c_{xi}^2 + c_{zi}^2 = a^2$, the circle and sphere are tangent and the manipulator is in a singular location. If $|c_{yi}| > b$, the circle and sphere do not intersect and there are no real solutions.

3.6.3 Direct Kinematics

For the direct kinematics, the input joint angles θ_{11} , θ_{12} , and θ_{13} are given and the problem is to find the position of the moving platform. First, we consider the surface composed of all possible positions of P for limb i with a given θ_{1i} . The surface is a sphere centered at B'_i , which is located at a distance h from B_i in the direction of $\overline{C_i P}$. Next, we consider all three limbs. Point P must fall simultaneously on the three spheres created by a sweep of P for each limb. The intersections of these three spheres represent the solutions to the direct kinematics problem. In the generic case, there are two solutions, since the intersection of two spheres forms a circle, which is generally intersected by the third sphere in two locations. Four cases are possible:

1. *Generic solution.* The two solutions are realized at the intersection of three spheres.
2. *Singular solution.* One sphere is tangent to the circle of intersection of the other two spheres. Hence there is only one solution possible.
3. *Singular solution.* The centers of any two spheres coincide, resulting in an infinite number of solutions. This is an unlikely configuration for most practical embodiments of the manipulator, except for the situation when $\theta_{11} = \theta_{12} = \theta_{13} = \pi/2$ and $r = h$.
4. *No solution.* The three spheres do not intersect.

Algebraically, the solution is found by writing the equations that describe the three spheres and then solving those equations for the points of intersection. Substituting c_{xi} , c_{yi} , and c_{zi} from Eq. (3.45) into (3.44) and rearranging, we obtain

$$\begin{bmatrix} bs\theta_{3i}c(\theta_{1i} + \theta_{2i}) \\ bc\theta_{3i} \\ bs\theta_{3i}s(\theta_{1i} + \theta_{2i}) \end{bmatrix} = \begin{bmatrix} c\phi_i & s\phi_i & 0 \\ -s\phi_i & c\phi_i & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} + \begin{bmatrix} -ac\theta_{1i} - r + h \\ 0 \\ -as\theta_{1i} \end{bmatrix}. \quad (3.49)$$

An equation for the sphere swept by point P for limb i is obtained by summing the squares of the three components of Eq. (3.49):

$$\begin{aligned} b^2 = & p_x^2 + p_y^2 + p_z^2 - 2(p_x c\phi_i + p_y s\phi_i)(ac\theta_{1i} + r - h) \\ & - 2p_z as\theta_{1i} + (ac\theta_{1i} + r - h)^2 + a^2 s^2 \theta_{1i} \end{aligned} \quad (3.50)$$

for $i = 1, 2$, and 3 .

The plane that contains the circle of intersection created by the spheres of limb 1 and limb j is found by subtracting Eq. (3.50) for $i = 1$ from Eq. (3.50) for $i = j$:

$$e_{1j}p_x + e_{2j}p_y + e_{3j}p_z + e_{4j} = 0 \quad \text{for } j = 2, 3, \quad (3.51)$$

where

$$\begin{aligned} e_{1j} &= 2c\phi_j(ac\theta_{1j} + r - h) - 2c\phi_1(ac\theta_{11} + r - h), \\ e_{2j} &= 2s\phi_j(ac\theta_{1j} + r - h) - 2s\phi_1(ac\theta_{11} + r - h), \\ e_{3j} &= 2as\theta_{1j} - 2as\theta_{11}, \\ e_{4j} &= (ac\theta_{11} + r - h)^2 + a^2 s^2 \theta_{11} - (ac\theta_{1j} + r - h)^2 - a^2 s^2 \theta_{1j}. \end{aligned}$$

Equation (3.51) written twice for $j = 2$ and 3 results in a system of two linearly independent equations as long as the centers of the spheres are not collinear. This system of equations defines a line that contains point P if there are real solutions. The intersections of this line with one of the spheres described by Eq. (3.50) solve the direct kinematics problem. In this case, solving Eq. (3.51) for p_y and p_z in terms of p_x and then substituting the resulting expressions into Eq. (3.50) for $i = 1$ yields

$$k_0 p_x^2 + k_1 p_x + k_2 = 0. \quad (3.52)$$

The coefficients of the quadratic are

$$\begin{aligned} k_0 &= 1 + \frac{l_1^2}{l_2^2} + \frac{l_4^2}{l_2^2}, \\ k_1 &= \frac{2l_0 l_1}{l_2^2} + \frac{2l_3 l_4}{l_2^2} - 2l_5 c\phi_1 - \frac{2l_5 l_1}{l_2} s\phi_1 - \frac{2al_4}{l_2} s\theta_{11}, \\ k_2 &= l_5^2 - b^2 + \frac{l_0^2}{l_2^2} + \frac{l_3^2}{l_2^2} + a^2 s^2 \theta_{11} - \frac{2l_0 l_5}{l_2} s\phi_1 - \frac{2al_3}{l_2} s\theta_{11}, \end{aligned}$$

where

$$\begin{aligned} l_0 &= e_{32}e_{43} - e_{33}e_{42}, \\ l_1 &= e_{13}e_{32} - e_{12}e_{33}, \\ l_2 &= e_{22}e_{33} - e_{23}e_{32}, \\ l_3 &= e_{23}e_{42} - e_{22}e_{43}, \\ l_4 &= e_{12}e_{23} - e_{13}e_{22}, \\ l_5 &= ac\theta_{11} + r - h, \end{aligned}$$

and where e_{12} , e_{13} , e_{22} , e_{23} , e_{32} , e_{33} , e_{42} , and e_{43} are as defined in Eq. (3.51).

The following four cases are possible:

1. If $k_1^2 - 4k_0k_2 > 0$, two solutions are realized where the circle created by the intersection of two spheres is intersected by the third sphere in two places. The two manipulator postures form a mirror image of each other about a plane defined by the centers of the three spheres.
2. If $k_1^2 - 4k_0k_2 = 0$, the circle created by the intersection of two spheres is tangent to the third sphere, resulting in one real solution.

3. If $k_1^2 - 4k_0k_2 < 0$, the three spheres do not intersect and there are no real solutions.
4. If the system of equations produced by Eq. (3.51) for $j = 2$ and 3 are linearly dependent, the centers of the spheres are collinear, resulting in either an infinite number of solutions if the centers of the spheres are coincident or no solutions if they are not coincident.

Once p_x is found, the values for p_y and p_z are found by back substitution into Eq. (3.51).

In summary, a parallel manipulator with 3 translational degrees of freedom is described. The general design of the manipulator is discussed. Closed-formed solutions are developed for both the direct and inverse kinematics. These solutions demonstrate that, in general, there are two possible poses for the forward kinematics and two possible poses for each limb for the inverse kinematics. A geometric approach to both the forward and inverse kinematics problems is also considered that provides some insights into the nature of the inverse and forward kinematics of this manipulator.

3.7 POSITION ANALYSIS OF A SPATIAL 3RPS PARALLEL MANIPULATOR

Figure 3.13 shows a spatial 3-dof, 3RPS parallel manipulator (Lee and Shah, 1987). Three identical limbs connect the moving platform at points B_i by spherical joints to the fixed base at points A_i by revolute joints. Each limb consists of an upper and a lower member connected by a prismatic joint. These three prismatic joints are used as the inputs to the manipulator. Overall, there are eight links, three revolute joints, three prismatic joints, and three spherical joints. Hence the degrees of freedom of the mechanism is

$$F = \lambda(n - j - 1) + \sum_i f_i = 6(8 - 9 - 1) + (3 + 3 + 9) = 3. \quad (3.53)$$

3.7.1 Geometry of the Manipulator

For the purpose of analysis, two Cartesian coordinate systems $A(x, y, z)$ and $B(u, v, w)$ are attached to the fixed base and moving platform, respectively, as shown in Fig. 3.13. The following assumptions are made. Points A_1, A_2 , and A_3 lie on the x - y plane and B_1, B_2 , and B_3 lie on the u - v plane. As shown in Fig. 3.14, the origin O of the fixed coordinate system is located at the centroid of $\Delta A_1A_2A_3$ and the x -axis points in the direction of $\overline{OA_1}$. Similarly, the origin P of the moving coordinate system is located at the

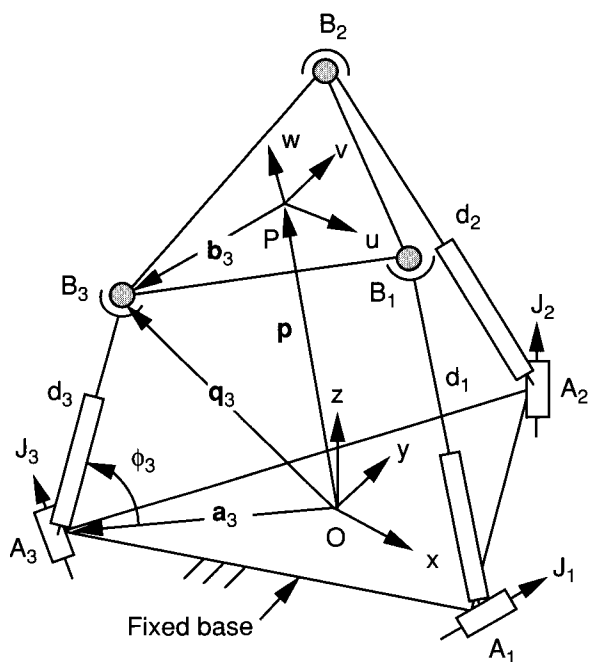
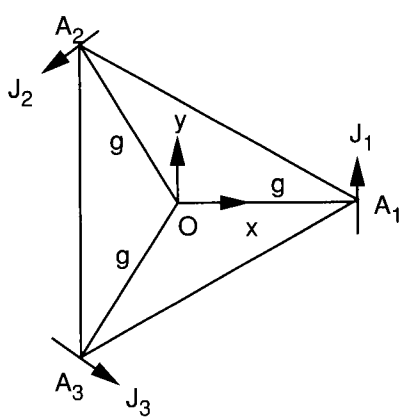
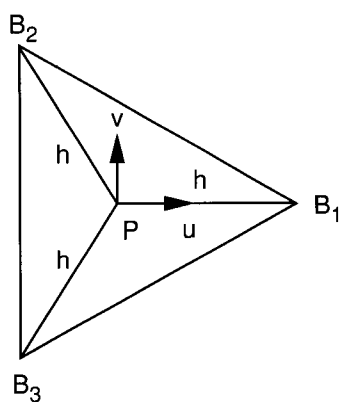


FIGURE 3.13. Spatial 3-dof, 3RPS parallel manipulator.



(a) Fixed base



(b) Moving platform

FIGURE 3.14. Top views of the 3RPS parallel manipulator.

centroid of $\Delta B_1 B_2 B_3$ and the u -axis points in the direction of $\overline{PB_1}$. Both $\Delta A_1 A_2 A_3$ and $\Delta B_1 B_2 B_3$ are equilateral triangles with $|OA_1| = |OA_2| = |OA_3| = g$ and $|PB_1| = |PB_2| = |PB_3| = h$. Furthermore, the axis of each revolute joint, J_i , lies on the x - y plane and is perpendicular to the vector $\overline{OA_i}$.

The transformation from the moving platform to the fixed base can be described by a position vector $\mathbf{p} = \overline{OP}$, and a 3×3 rotation matrix ${}^A R_B$. Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be three unit vectors defined along the u , v , and w axes of the moving coordinate system B , respectively; then the rotation matrix can be expressed in terms of the direction cosines of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ as

$${}^A R_B = \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix}. \quad (3.54)$$

We note that the elements of ${}^A R_B$ must satisfy the following orthogonal conditions:

$$u_x^2 + u_y^2 + u_z^2 = 1, \quad (3.55)$$

$$v_x^2 + v_y^2 + v_z^2 = 1, \quad (3.56)$$

$$w_x^2 + w_y^2 + w_z^2 = 1, \quad (3.57)$$

$$u_x v_x + u_y v_y + u_z v_z = 0, \quad (3.58)$$

$$u_x w_x + u_y w_y + u_z w_z = 0, \quad (3.59)$$

$$v_x w_x + v_y w_y + v_z w_z = 0. \quad (3.60)$$

Let $\mathbf{a}_i$ and ${}^B \mathbf{b}_i$ be the position vectors of points A_i and B_i in the coordinate systems A and B , respectively. Then the coordinates of A_i and B_i are given by

$$\mathbf{a}_1 = [g, 0, 0]^T, \quad (3.61)$$

$$\mathbf{a}_2 = \left[-\frac{1}{2}g, \frac{\sqrt{3}}{2}g, 0 \right]^T, \quad (3.62)$$

$$\mathbf{a}_3 = \left[-\frac{1}{2}g, -\frac{\sqrt{3}}{2}g, 0 \right]^T, \quad (3.63)$$

$${}^B \mathbf{b}_1 = [h, 0, 0]^T, \quad (3.64)$$

$${}^B\mathbf{b}_2 = \begin{bmatrix} -\frac{1}{2}h, \frac{\sqrt{3}}{2}h, 0 \end{bmatrix}^T, \quad (3.65)$$

$${}^B\mathbf{b}_3 = \begin{bmatrix} -\frac{1}{2}h, -\frac{\sqrt{3}}{2}h, 0 \end{bmatrix}^T. \quad (3.66)$$

The position vector $\mathbf{q}_i$ of B_i with respect to the fixed coordinate system is obtained by the following transformation:

$$\mathbf{q}_i = \mathbf{p} + {}^A R_B {}^B \mathbf{b}_i. \quad (3.67)$$

Substituting Eqs. (3.54) and (3.64) through (3.66) into (3.67) yields

$$\mathbf{q}_1 = \begin{bmatrix} p_x + hu_x \\ p_y + hu_y \\ p_z + hu_z \end{bmatrix}, \quad (3.68)$$

$$\mathbf{q}_2 = \begin{bmatrix} p_x - \frac{1}{2}hu_x + \frac{\sqrt{3}}{2}hv_x \\ p_y - \frac{1}{2}hu_y + \frac{\sqrt{3}}{2}hv_y \\ p_z - \frac{1}{2}hu_z + \frac{\sqrt{3}}{2}hv_z \end{bmatrix}, \quad (3.69)$$

$$\mathbf{q}_3 = \begin{bmatrix} p_x - \frac{1}{2}hu_x - \frac{\sqrt{3}}{2}hv_x \\ p_y - \frac{1}{2}hu_y - \frac{\sqrt{3}}{2}hv_y \\ p_z - \frac{1}{2}hu_z - \frac{\sqrt{3}}{2}hv_z \end{bmatrix}. \quad (3.70)$$

3.7.2 Constraints Imposed by the Revolute Joints

Since the each limb is connected to the fixed base by a revolute joint located at A_i , its motion is constrained in one of the following three planes:

$$q_{1y} = 0 \quad \text{for } i = 1, \quad (3.71)$$

$$q_{2y} = -\sqrt{3}q_{2x} \quad \text{for } i = 2, \quad (3.72)$$

$$q_{3y} = +\sqrt{3}q_{3x} \quad \text{for } i = 3. \quad (3.73)$$

Substituting the y -component of $\mathbf{q}_i$ from Eqs. (3.68) through (3.70) into the three equations above, we obtain

$$p_y + hu_y = 0, \quad (3.74)$$

$$p_y - \frac{1}{2}hu_y + \frac{\sqrt{3}}{2}hv_y = -\sqrt{3}\left(p_x - \frac{1}{2}hu_x + \frac{\sqrt{3}}{2}hv_x\right), \quad (3.75)$$

$$p_y - \frac{1}{2}hu_y - \frac{\sqrt{3}}{2}hv_y = \sqrt{3}\left(p_x - \frac{1}{2}hu_x - \frac{\sqrt{3}}{2}hv_x\right). \quad (3.76)$$

Equations (3.75) and (3.76) can be simplified further. Subtracting the sum of Eqs. (3.75) and (3.76) from $2 \times (3.74)$, we obtain

$$v_x = u_y. \quad (3.77)$$

Subtracting Eq. (3.76) from (3.75), we obtain

$$p_x = \frac{1}{2}h(u_x - v_y). \quad (3.78)$$

Hence Eqs. (3.74), (3.77), and (3.78) impose three constraints on the motion of the moving platform.

3.7.3 Length of Limb i

The length of a limb, d_i , shown in Fig. 3.13, is given by

$$d_i^2 = [\mathbf{q}_i - \mathbf{a}_i]^T [\mathbf{q}_i - \mathbf{a}_i] \quad \text{for } i = 1, 2, 3. \quad (3.79)$$

Substituting Eqs. (3.61) through (3.63) and (3.68) through (3.70) into (3.79) yields

$$\begin{aligned} d_1^2 = & p_x^2 + p_y^2 + p_z^2 + 2h(p_x u_x + p_y u_y + p_z u_z) \\ & - 2gp_x - 2ghu_x + g^2 + h^2, \end{aligned} \quad (3.80)$$

$$\begin{aligned} d_2^2 = & p_x^2 + p_y^2 + p_z^2 - h(p_x u_x + p_y u_y + p_z u_z) \\ & + \sqrt{3}h(p_x v_x + p_y v_y + p_z v_z) + g(p_x - \sqrt{3}p_y) \\ & - \frac{1}{2}gh(u_x - \sqrt{3}u_y) + \frac{1}{2}gh(\sqrt{3}v_x - 3v_y) + g^2 + h^2, \end{aligned} \quad (3.81)$$

$$\begin{aligned}
d_3^2 = & p_x^2 + p_y^2 + p_z^2 - h(p_x u_x + p_y u_y + p_z u_z) \\
& - \sqrt{3}h(p_x v_x + p_y v_y + p_z v_z) + g(p_x + \sqrt{3}p_y) \\
& - \frac{1}{2}gh(u_x + \sqrt{3}u_y) - \frac{1}{2}gh(\sqrt{3}v_x + 3v_y) + g^2 + h^2. \quad (3.82)
\end{aligned}$$

Equations (3.80), (3.81), and (3.82) relate the limb lengths to the moving platform location.

3.7.4 Inverse Kinematics

The inverse kinematic problem is: *Given the location of the moving platform, find the limb lengths d_1 , d_2 , and d_3 .* Since the manipulator has only 3 degrees of freedom, the position and orientation must be specified in accordance with the constraints imposed by the revolute joints. Equation (3.77) imposes a constraint on the orientation of the moving platform, while Eqs. (3.74) and (3.78) relate the x and y components of $\mathbf{p}$ to the orientation of the moving platform. Hence only three of the 12 parameters in ${}^A R_B$ and $\mathbf{p}$ can be specified arbitrarily. In any case, the z -component of $\mathbf{p}$ must be specified, since it does not appear in the constraint equations, while the other two parameters can be chosen from either the position vector $\mathbf{p}$ or the three unit vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. Once three parameters are chosen, the remaining parameters are determined from Eqs. (3.74), (3.77), and (3.78) and the orthogonal conditions. It can be shown that if the three components of $\mathbf{p}$ are chosen as the independent parameters, there are eight corresponding platform orientations (Song and Zhang, 1995). If the roll and pitch angles of the Euler roll-pitch-yaw angles and p_z are chosen as the independent parameters, there are two corresponding platform locations. Once the position vector and the rotation matrix of the moving platform are known, the limb lengths can be computed directly from Eqs. (3.80), (3.81), and (3.82).

3.7.5 Direct Kinematics

The direct kinematic problem is: *Given the limb lengths d_1 , d_2 , and d_3 , find the location of the moving platform.* Equations (3.55), (3.56), (3.58), (3.74), (3.77), (3.78), and (3.80) through (3.82) form a system of nine polynomial equations in nine unknowns: p_x , p_y , p_z , u_x , u_y , u_z , v_x , v_y , and v_z . Equations (3.55), (3.56), (3.58), (3.80), (3.81), and (3.82) are second-degree polynomials, while Eqs. (3.74), (3.77), and (3.78) are linear equations. Two linear equations can be obtained by manipulating Eqs. (3.80), (3.81), and (3.82) algebraically. This results in a new system with four second-degree polynomials and five linear equations. Since the traditional Bezout number is $2^4 = 16$, the system of equations has at most 16 solutions.

In what follows we introduce the tilt angles of the limbs, ϕ_i , as intermediate variables and show how to use these angles to reduce the problem to a 16th-degree polynomial. Let ϕ_i be measured from line A_iO to A_iB_i as shown in Fig. 3.13. Then the position vector of point B_i with respect to the fixed coordinate system can be written as

$$\mathbf{q}_1 = \begin{bmatrix} g - d_1 c\phi_1 \\ 0 \\ d_1 s\phi_1 \end{bmatrix}, \quad (3.83)$$

$$\mathbf{q}_2 = \begin{bmatrix} -\frac{1}{2}(g - d_2 c\phi_2) \\ +\frac{\sqrt{3}}{2}(g - d_2 c\phi_2) \\ d_2 s\phi_2 \end{bmatrix}, \quad (3.84)$$

$$\mathbf{q}_3 = \begin{bmatrix} -\frac{1}{2}(g - d_3 c\phi_3) \\ -\frac{\sqrt{3}}{2}(g - d_3 c\phi_3) \\ d_3 s\phi_3 \end{bmatrix}. \quad (3.85)$$

We note that once the limb lengths are given, the resulting mechanism is equivalent to a $3RS$ spatial mechanism with zero degrees of freedom. Hence the position and orientation of the mechanism can be obtained by equating $|B_i B_{i+1}| = \sqrt{3}h$; that is,

$$[\mathbf{q}_i - \mathbf{q}_{i+1}]^T [\mathbf{q}_i - \mathbf{q}_{i+1}] - 3h^2 = 0 \quad \text{for } i = 1, 2, 3 \quad (\text{modulo } 3). \quad (3.86)$$

Note that the subscripts in Eq. (3.86) are cyclic of modulo 3. Thus if $i = 3$, then $i + 1$ is equal to 1. Substituting Eqs. (3.83) through (3.85) into (3.86) yields

$$e_{1i} c\phi_i c\phi_{i+1} + e_{2i} s\phi_i s\phi_{i+1} + e_{3i} c\phi_i + e_{4i} c\phi_{i+1} + e_{5i} = 0, \quad (3.87)$$

for $i = 1, 2, 3$ (modulo 3), where

$$\begin{aligned} e_{1i} &= d_i d_{i+1}, \\ e_{2i} &= -2d_i d_{i+1}, \\ e_{3i} &= -3gd_i, \\ e_{4i} &= -3gd_{i+1}, \\ e_{5i} &= 3g^2 + d_i^2 + d_{i+1}^2 - 3h^2. \end{aligned}$$

We now convert Eqs. (3.87) to a system of polynomial equations in order to eliminate two unknown variables. Substituting the trigonometric identities

$$s\phi_i = \frac{2t_i}{1+t_i^2} \quad \text{and} \quad c\phi_i = \frac{1-t_i^2}{1+t_i^2}, \quad \text{where} \quad t_i = \tan \frac{\phi_i}{2}$$

into Eq. (3.87) yields three fourth-degree polynomials in t_1 , t_2 , and t_3 :

$$\tilde{e}_{1i}t_i^2t_{i+1}^2 + \tilde{e}_{2i}t_i^2 + \tilde{e}_{3i}t_{i+1}^2 + \tilde{e}_{4i}t_it_{i+1} + \tilde{e}_{5i} = 0 \quad \text{for } i = 1, 2, 3 \text{ (modulo 3)} \quad (3.88)$$

where

$$\tilde{e}_{1i} = e_{1i} - e_{3i} - e_{4i} + e_{5i},$$

$$\tilde{e}_{2i} = -e_{1i} - e_{3i} + e_{4i} + e_{5i},$$

$$\tilde{e}_{3i} = -e_{1i} + e_{3i} - e_{4i} + e_{5i},$$

$$\tilde{e}_{4i} = 4e_{2i},$$

$$\tilde{e}_{5i} = e_{1i} + e_{3i} + e_{4i} + e_{5i}.$$

Although, the traditional 1-homogeneous Bezout number is $4^3 = 64$, the 3-homogeneous Bezout number is 16. To prove this, we arrange the variables into three groups:

group 1: $[t_1]$, group 2: $[t_2]$, and group 3: $[t_3]$.

The degrees of the three equations in the variables of each group are listed in Table 3.2. Following the procedure outlined in Appendix A, we compute the product

$$\prod_{i=1}^3 \left(\sum_{j=1}^3 d_{ij} \beta_j \right) = (2\beta_1 + 2\beta_2)(2\beta_2 + 2\beta_3)(2\beta_1 + 2\beta_3),$$

where d_{ij} is taken from the (i, j) element of Table 3.2. Since each group contains only one variable, the 3-homogeneous Bezout number, determined by the coefficient of the term $\beta_1\beta_2\beta_3$, is 16. Hence the system of equations has at most 16 solutions.

TABLE 3.2. Degrees of the Equations

Equation	Group 1	Group 2	Group 3
1, Eq. (3.88) for $i = 1$	2	2	0
2, Eq. (3.88) for $i = 2$	0	2	2
3, Eq. (3.88) for $i = 3$	2	0	2

In what follows, we apply the Sylvester dialytic elimination method to reduce the system of equations to a 16th-degree polynomial. See Appendix B for a brief description of the method.

(a) Elimination of t_3 . To eliminate t_3 , we write Eq. (3.88) for $i = 2$ and 3 as two second-degree polynomials in t_3 :

$$At_3^2 + Bt_3 + C = 0, \quad (3.89)$$

$$Dt_3^2 + Et_3 + F = 0, \quad (3.90)$$

where A , B , and C are second-degree polynomials in t_2 whose coefficients can be derived from Eq. (3.88) for $i = 2$, where as D , E , and F are second-degree polynomials in t_1 whose coefficients can be derived from Eq. (3.88) for $i = 3$.

Multiplying Eq. (3.89) by D and Eq. (3.90) by A , and subtracting, we obtain

$$(AE - BD)t_3 + (AF - CD) = 0. \quad (3.91)$$

Multiplying Eq. (3.89) by F and Eq. (3.90) by C and subtracting, we obtain

$$(CD - AF)t_3 + (CE - BF) = 0. \quad (3.92)$$

Equations (3.91) and (3.92) constitute a system of two equations linear in t_3 and 1. The following eliminant is obtained by equating the determinant of the coefficient matrix to zero.

$$(AF - CD)^2 + (AE - BD)(CE - BF) = 0. \quad (3.93)$$

(b) Elimination of t_2 . For the purpose of eliminating t_2 , we write Eq. (3.93) in the form

$$Lt_2^4 + Mt_2^3 + Nt_2^2 + Pt_2 + Q = 0, \quad (3.94)$$

where L , M , N , P , and Q are fourth-degree polynomials in t_1 . We also write Eq. (3.88) for $i = 1$ in the form

$$Gt_2^2 + Ht_2 + I = 0, \quad (3.95)$$

where G , H , and I are second-degree polynomials in t_1 .

We now eliminate the unknown t_2 from Eqs. (3.94) and (3.95) as follows. Multiplying Eq. (3.94) by G , Eq. (3.95) by Lt_2^2 , and subtracting, we obtain

$$(HL - GM)t_2^3 + (IL - GN)t_2^2 - GPt_2 - GQ = 0. \quad (3.96)$$

Multiplying Eq. (3.94) by $Gt_2 + H$, Eq. (3.95) by $Lt_2^3 + Mt_2^2$, and subtracting, we obtain

$$(GN - LI)t_2^3 + (GP + HN - MI)t_2^2 + (GQ + HP)t_2 + HQ = 0. \quad (3.97)$$

Multiplying Eq. (3.95) by t_2 , we obtain

$$Gt_2^3 + Ht_2^2 + It_2 = 0. \quad (3.98)$$

We may consider Eqs. (3.95) through (3.98) as four linear homogeneous equations in the four variables t_2^3 , t_2^2 , t_2 , and 1. Vanishing of their eliminant yields

$$\begin{vmatrix} HL - GM & IL - GN & -GP & -GQ \\ GN - LI & GP + HN - MI & GQ + HP & HQ \\ G & H & I & 0 \\ 0 & G & H & I \end{vmatrix} = 0. \quad (3.99)$$

Expanding Eq. (3.99) results in an eighth-degree polynomial in the square of t_1 . It follows that there are at most eight pairs of solutions for t_1 , one being the negative of the other.

(c) Location of the Moving Platform. Once ϕ_1 is found, ϕ_2 and ϕ_3 can be solved by back substitution using Eqs. (3.95) and (3.89). Next, the position vectors $\mathbf{q}_1$, $\mathbf{q}_2$, and $\mathbf{q}_3$ are found by substituting the values of ϕ_i into Eqs. (3.83), (3.84), and (3.85), respectively. Since the ball joints are located at the vertices of the equilateral triangle, the position vector of the moving platform, $\mathbf{p}$, is obtained as

$$\mathbf{p} = \frac{1}{3}(\mathbf{q}_1 + \mathbf{q}_2 + \mathbf{q}_3). \quad (3.100)$$

Finally, the orientation of the moving platform is found by solving Eqs. (3.74), (3.77), and (3.78) along with the orthogonal conditions.

In summary, there are at most 16 solutions for the direct kinematics problem. These 16 solutions form eight pairs of manipulator postures, one being the mirror image of another about the fixed plane $A_1A_2A_3$.

3.8 POSITION ANALYSIS OF A GENERAL STEWART-GOUGH PLATFORM

Figure 3.15 shows a spatial 6-dof, 6SPS parallel manipulator known as a Stewart-Gough platform. Six identical limbs connect the moving platform to the fixed base by spherical joints at points B_i and A_i , $i = 1, 2, \dots, 6$,

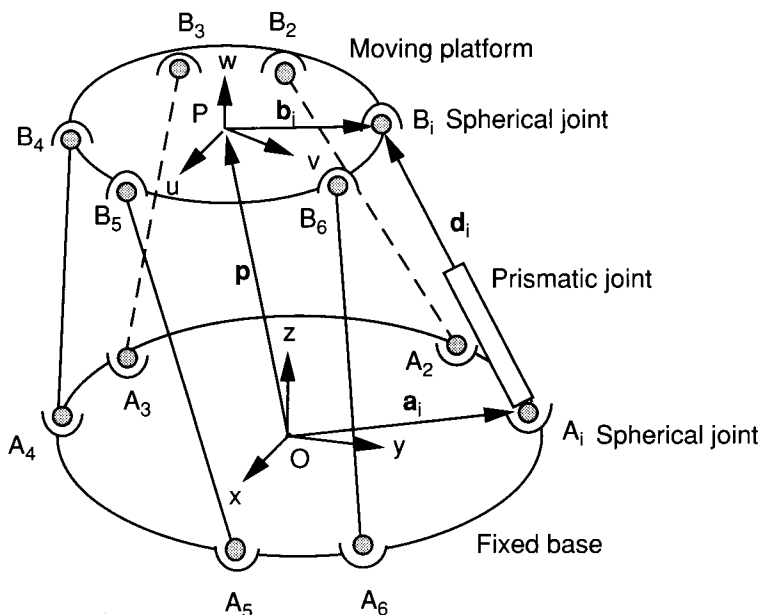


FIGURE 3.15. Spatial 6-dof, 6SPS parallel manipulator.

respectively. Each limb consists of an upper member and a lower member connected by a prismatic joint. Ball screws or hydraulic jacks can be used to vary the lengths of the prismatic joints and therefore to control the location of the moving platform.

Note that in Fig. 3.15, the attachment points A_i for $i = 1$ to 6 are sketched in a plane on the fixed base. Similarly, B_i for $i = 1$ to 6 are sketched in a plane on the moving platform. For a general Stewart platform, however, these attachment points do not necessarily lie on one plane. There are 14 links connected by 6 prismatic joints and 12 spherical joints. Hence the number of degrees of freedom of the mechanism is

$$F = \lambda(n - j - 1) + \sum_i f_i = 6(14 - 18 - 1) + (6 + 3 \times 12) = 12. \quad (3.101)$$

However, there are 6 passive degrees of freedom associated with the six SPS limbs. Therefore, the moving platform possesses 6 degrees of freedom. Note that an SPS limb can be replaced by an SPU limb without compromising the overall degrees of freedom of the mechanism.

Since the limbs are connected to the moving platform and the fixed base by spherical joints, no bending moments or twisting torques will transmit to

the limbs. The force acting on each limb is directed along the longitudinal axis of the limb. Consequently, these limbs can be made of hollow cylindrical rods to produce a light-weight, high-stiffness, high-speed manipulator.

3.8.1 Geometry of the Manipulator

For the purpose of analysis, two Cartesian coordinate systems, frames $A(x, y, z)$ and $B(u, v, w)$ as shown in Fig. 3.15, are attached to the fixed base and moving platform, respectively. The transformation from the moving platform to the fixed base can be described by the position vector $\mathbf{p}$ of the centroid P and the rotation matrix ${}^A R_B$ of the moving platform. Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be three unit vectors defined along the u , v , and w axes of the moving coordinate system; then the rotation matrix can be written as

$${}^A R_B = \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix}. \quad (3.102)$$

Note that the elements of ${}^A R_B$ must satisfy the following orthogonal conditions:

$$u_x^2 + u_y^2 + u_z^2 = 1, \quad (3.103)$$

$$v_x^2 + v_y^2 + v_z^2 = 1, \quad (3.104)$$

$$w_x^2 + w_y^2 + w_z^2 = 1, \quad (3.105)$$

$$u_x v_x + u_y v_y + u_z v_z = 0, \quad (3.106)$$

$$u_x w_x + u_y w_y + u_z w_z = 0, \quad (3.107)$$

$$v_x w_x + v_y w_y + v_z w_z = 0. \quad (3.108)$$

As shown in Fig. 3.15, let $\mathbf{a}_i = [a_{ix}, a_{iy}, a_{iz}]^T$ and ${}^B \mathbf{b}_i = [b_{iu}, b_{iv}, b_{iw}]^T$ be the position vectors of points A_i and B_i in the coordinate frames A and B , respectively. We can write a vector-loop equation for the i th limb of the manipulator as follows:

$$\overline{A_i B_i} = \mathbf{p} + {}^A R_B {}^B \mathbf{b}_i - \mathbf{a}_i. \quad (3.109)$$

The length of the i th limb is obtained by taking the dot product of the vector $\overline{A_i B_i}$ with itself:

$$d_i^2 = [\mathbf{p} + {}^A R_B {}^B \mathbf{b}_i - \mathbf{a}_i]^T [\mathbf{p} + {}^A R_B {}^B \mathbf{b}_i - \mathbf{a}_i], \quad \text{for } i = 1, 2, \dots, 6, \quad (3.110)$$

where d_i denotes the length of the i th limb. Expanding Eq. (3.110) yields

$$d_i^2 = \mathbf{p}^T \mathbf{p} + [{}^B \mathbf{b}_i]^T [{}^B \mathbf{b}_i] + \mathbf{a}_i^T \mathbf{a}_i + 2\mathbf{p}^T [{}^A R_B {}^B \mathbf{b}_i] - 2\mathbf{p}^T \mathbf{a}_i - 2[{}^A R_B {}^B \mathbf{b}_i]^T \mathbf{a}_i. \quad (3.111)$$

Equation (3.111) written six times, once for each $i = 1, 2, \dots, 6$, yields six equations describing the location of the moving platform with respect to the fixed base. Note that ${}^B \mathbf{b}_i$ and $\mathbf{a}_i$ are constant vectors defined by the geometry of the manipulator.

3.8.2 Inverse Kinematics

For the inverse kinematics problem, the position vector $\mathbf{p}$ and rotation matrix ${}^A R_B$ of frame B with respect to A are given and the limb lengths d_i , $i = 1, 2, \dots, 6$, are to be found. The solution is very straightforward. Taking the square root of Eq. (3.111) we obtain

$$d_i = \pm \sqrt{\mathbf{p}^T \mathbf{p} + {}^B \mathbf{b}_i^T {}^B \mathbf{b}_i + \mathbf{a}_i^T \mathbf{a}_i + 2\mathbf{p}^T {}^A R_B {}^B \mathbf{b}_i - 2\mathbf{p}^T \mathbf{a}_i - 2[{}^A R_B {}^B \mathbf{b}_i]^T \mathbf{a}_i} \quad (3.112)$$

for $i = 1, 2, \dots, 6$. Hence, corresponding to each given location of the moving platform, there are generally two possible solutions for each limb. However, the negative limb length is physically not feasible. When the solution of d_i becomes a complex number, the location of the moving platform is not reachable.

3.8.3 Direct Kinematics

For the direct kinematics problem, the limb lengths d_i , for $i = 1, 2, \dots, 6$, are given, and the position vector $\mathbf{p}$ and rotation matrix ${}^A R_B$ of the moving platform are to be found. The position vector contains three scalar unknowns, while the rotation matrix contains nine scalar unknowns. However, the nine scalar unknowns in ${}^A R_B$ are related by the six orthogonal conditions given by Eqs. (3.103) through (3.108). Without losing generality, we may assume that

1. The origin O of the fixed frame is located at the center of the fixed spherical joint A_1 .
2. The origin P of the moving frame is located at the center of the moving spherical joint B_1 .

Based on the assumptions above, we have $a_{1x} = a_{1y} = a_{1z} = 0$ and $b_{1u} = b_{1v} = b_{1w} = 0$. Hence Eq. (3.111) for $i = 1$ reduces to

$$d_1^2 = p_x^2 + p_y^2 + p_z^2. \quad (3.113)$$

Expanding (3.111) for $i = 2, 3, \dots, 6$, and then subtracting Eq. (3.113) from each of the resulting equations yields

$$\begin{aligned}
 & b_{iu}(p_x u_x + p_y u_y + p_z u_z) + b_{iv}(p_x v_x + p_y v_y + p_z v_z) \\
 & + b_{iw}(p_x w_x + p_y w_y + p_z w_z) - a_{ix} p_x - a_{iy} p_y - a_{iz} p_z \\
 & - b_{iu}(a_{ix} u_x + a_{iy} u_y + a_{iz} u_z) - b_{iv}(a_{ix} v_x + a_{iy} v_y + a_{iz} v_z) \\
 & - b_{iw}(a_{ix} w_x + a_{iy} w_y + a_{iz} w_z) + k_i = 0 \quad \text{for } i = 2, 3, \dots, 6,
 \end{aligned}
 \tag{3.114}$$

where $k_i = (a_{ix}^2 + a_{iy}^2 + a_{iz}^2 + b_{iu}^2 + b_{iv}^2 + b_{iw}^2 + d_1^2 - d_i^2)/2$.

Equations (3.113) and (3.114) for $i = 2$ to 6, together with the six orthogonal conditions, yields 12 equations in 12 unknowns. These 12 equations can be solved for the direct kinematics problem of the general Stewart-Gough platform. The equations are highly nonlinear and extremely difficult to solve. Since each equation is of second degree, the traditional 1-homogeneous Bezout number is $2^{12} = 4096$, which contains many extraneous solutions.

For the 2-homogeneous formulation, we arrange the variables into two groups:

group 1: $[p_x, p_y, p_z]$ and group 2: $[u_x, u_y, u_z, v_x, v_y, v_z, w_x, w_y, w_z]$.

The degrees of the 12 equations in the variables of each group are listed in Table 3.3. Following the procedure outlined in Appendix A, we form the

TABLE 3.3. Degrees of the 12 Equations

Equation	Group 1	Group 2
1, Eq. (3.113)	2	0
2, Eq. (3.114) for $i = 2$	1	1
3, Eq. (3.114) for $i = 3$	1	1
4, Eq. (3.114) for $i = 4$	1	1
5, Eq. (3.114) for $i = 5$	1	1
6, Eq. (3.114) for $i = 6$	1	1
7, Eq. (3.103)	0	2
8, Eq. (3.104)	0	2
9, Eq. (3.105)	0	2
10, Eq. (3.106)	0	2
11, Eq. (3.107)	0	2
12, Eq. (3.108)	0	2

product

$$\prod_{i=1}^{12} \left(\sum_{j=1}^2 d_{ij} \beta_j \right) = (2\beta_1)(\beta_1 + \beta_2)^5 (2\beta_2)^6,$$

where d_{ij} is taken from the (i, j) element of Table 3.3.

Since the first group contains three variables and the second group contains nine variables, the 2-homogeneous Bezout number is given by the coefficient of the $\beta_1^3 \beta_2^9$ term in the polynomial above and is found to be 1280. Hence the system of equations has at most 1280 solutions. Although the 2-homogeneous Bezout number is much lower than the 1-homogeneous Bezout number, it still contains numerous extraneous solutions. Raghavan (1993) applied the continuation method and found that the general Stewart–Gough platform has 40 direct kinematics solutions.

3.9 POSITION ANALYSIS OF A NEARLY GENERAL STEWART–GOUGH PLATFORM

In this section we study the direct kinematics of a nearly general Stewart–Gough platform by introducing the following additional assumptions (Zhang and Song, 1994):

1. The six spherical joints on the moving platform lie on the u – v plane.
2. The six spherical joints on the fixed platform lie on the x – y plane.

Based on the assumptions above, we have $a_{iz} = b_{iw} = 0$ for $i = 1, 2, \dots, 6$. Hence Eq. (3.114) can be further reduced to

$$\begin{aligned} & b_{iu} \xi_1 + b_{iv} \xi_2 - a_{ix} p_x - a_{iy} p_y - a_{ix} b_{iu} u_x - a_{iy} b_{iu} u_y \\ & - a_{ix} b_{iv} v_x - a_{iy} b_{iv} v_y + k_i = 0 \quad \text{for } i = 2, 3, \dots, 6. \end{aligned} \quad (3.115)$$

where $k_i = (a_{ix}^2 + a_{iy}^2 + b_{iu}^2 + b_{iv}^2 - d_i^2 + d_1^2)/2$ and where the variables ξ_1 and ξ_2 are defined as follows:

$$\xi_1 = p_x u_x + p_y u_y + p_z u_z, \quad (3.116)$$

$$\xi_2 = p_x v_x + p_y v_y + p_z v_z. \quad (3.117)$$

We note that the unit vector $\mathbf{w}$ disappears from Eq. (3.115). Hence only three of the six orthogonal conditions are needed for the analysis. By considering ξ_1 and ξ_2 as two intermediate variables, we have a system of 11

equations in 11 unknowns. The system of equations consists of Eqs. (3.103), (3.104), (3.106), (3.113), (3.115) for $i = 2$ to 6, (3.116), and (3.117). The unknowns are $p_x, p_y, p_z, u_x, u_y, u_z, v_x, v_y, v_z, \xi_1$, and ξ_2 . Equations (3.115) for $i = 2$ to 6 are linear, while the remaining equations are second-degree polynomials.

In what follows, we derive six polynomial equations in three unknowns: u_x, u_y , and v_y . First, we rewrite Eqs. (3.103), (3.104), and (3.113) as a group, and (3.106), (3.116), and (3.117) as another group, as follows:

$$u_z^2 = 1 - u_x^2 - u_y^2, \quad (3.118)$$

$$v_z^2 = 1 - v_x^2 - v_y^2, \quad (3.119)$$

$$p_z^2 = d_1^2 - p_x^2 - p_y^2, \quad (3.120)$$

and

$$u_z v_z = -u_x v_x - u_y v_y, \quad (3.121)$$

$$p_z u_z = \xi_1 - p_x u_x - p_y u_y, \quad (3.122)$$

$$p_z v_z = \xi_2 - p_x v_x - p_y v_y. \quad (3.123)$$

Next, we substitute Eqs. (3.118) through (3.123) into the following six identities:

$$(u_z^2)(v_z^2) - (u_z v_z)^2 = 0, \quad (3.124)$$

$$(p_z^2)(u_z^2) - (p_z u_z)^2 = 0, \quad (3.125)$$

$$(p_z^2)(v_z^2) - (p_z v_z)^2 = 0, \quad (3.126)$$

$$(u_z v_z)(p_z^2) - (p_z u_z)(p_z v_z) = 0, \quad (3.127)$$

$$(p_z u_z)(v_z^2) - (u_z v_z)(p_z v_z) = 0, \quad (3.128)$$

$$(p_z v_z)(u_z^2) - (u_z v_z)(p_z u_z) = 0. \quad (3.129)$$

This results in six equations free of the variables u_z, v_z , and p_z .

Equations (3.115), $i = 2, 3, \dots, 6$, represent five linear equations in eight unknowns. Hence we can solve five unknowns in terms of the remaining three. For example, we can express ξ_1, ξ_2, p_x, p_y , and v_x in terms of u_x, u_y , and v_y as follows:

$$\xi_1 = e_{11}u_x + e_{12}u_y + e_{13}v_y + e_{14}, \quad (3.130)$$

$$\xi_2 = e_{21}u_x + e_{22}u_y + e_{23}v_y + e_{24}, \quad (3.131)$$

$$p_x = e_{31}u_x + e_{32}u_y + e_{33}v_y + e_{34}, \quad (3.132)$$

$$p_y = e_{41}u_x + e_{42}u_y + e_{43}v_y + e_{44}, \quad (3.133)$$

$$v_x = e_{51}u_x + e_{52}u_y + e_{53}v_y + e_{54}, \quad (3.134)$$

where $e_{i,j}$'s are constants that can be found by solving Eqs. (3.115) for $i = 2, 3, \dots, 6$.

Upon substitution of Eqs. (3.130) through (3.134) into Eqs. (3.124) through (3.129), we obtain six fourth-degree polynomials in three unknowns: u_x , u_y , and v_y . Any three of the six polynomials can be used to solve for the three unknowns. Since the traditional Bezout number is $4^3 = 64$, the system of equations has at most 64 solutions. Zhang and Song (1994) applied the Sylvester dialytic method to further reduce the six polynomial equations to a 20th-degree polynomial in one unknown and showed that after back substitution, there are at most 40 direct kinematics solutions. Closed-form solutions of this manipulator have also been derived by Wen and Liang (1994).

3.10 POSITION ANALYSIS OF A 3–3 STEWART–GOUGH PLATFORM

A major problem associated with the general Stewart–Gough platform manipulator is that its direct kinematics cannot be solved in a closed form, although it has been shown that there exist up to 40 direct kinematic solutions. To avoid the problems associated with the general Stewart–Gough platform, special configurations of the Stewart–Gough platform have been proposed. These special designs usually contains concentric spherical joints either at the moving platform or at the fixed base. Figure 3.16 shows a 3–3 Stewart–Gough platform whose six limbs meet in a pairwise fashion at three points each in the moving platform and the base. This type of special construction makes closed-form direct kinematics solutions feasible.

Figure 3.17 shows the top view of the 3–3 Stewart–Gough platform, where the moving platform is symbolically represented by three pairs of concentric spherical joints located at points B_1 , B_2 , and B_3 , and the fixed base is represented by another three pairs of concentric spherical joints located at points A_1 , A_2 , and A_3 . For the purpose of analysis, we attach a Cartesian coordinate system to the fixed base with its origin located at point A_1 , the x -axis pointing along line A_1A_2 , the z -axis perpendicular to the plane defined by the triangle $\Delta A_1A_2A_3$, and the y -axis defined by the right-hand screw rule as shown in Fig. 3.17.

For the direct kinematics problem, the limb lengths are given and the platform location is to be found. Because of the special configuration, once the limb lengths are given, three triangles $\Delta A_1A_2B_1$, $\Delta A_2A_3B_2$, and $\Delta A_3A_1B_3$

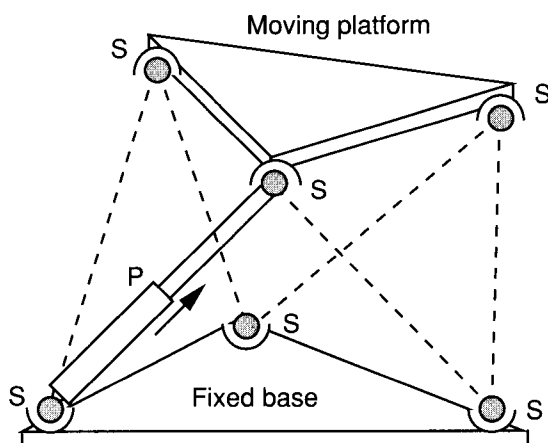


FIGURE 3.16. Spatial 6-dof, 3-3 Stewart-Gough platform.

can be uniquely determined. Hence the locus of B_i is constrained on a circle that is centered at and perpendicular to line $A_i A_{i+1}$. This constraint is equivalent to a simple RS (revolute-spherical) joint pair. Hence the overall mechanism is kinematically equivalent to a spatial $3RS$ mechanism with zero degrees of freedom, as shown in Fig. 3.18.

To perform the analysis, we first identify the center C_i and equivalent radius r_i of the circle on which point B_i lies. Then we express the coordinates of point B_i in terms of an angle ϕ_i which the triangle $\Delta A_i A_{i+1} B_i$ makes with

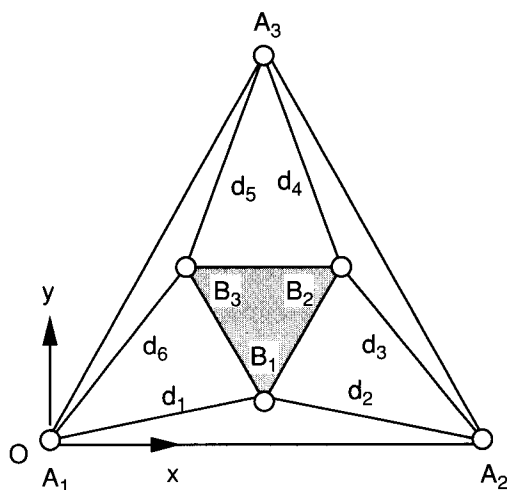


FIGURE 3.17. Top view of the 3-3 Stewart-Gough platform.

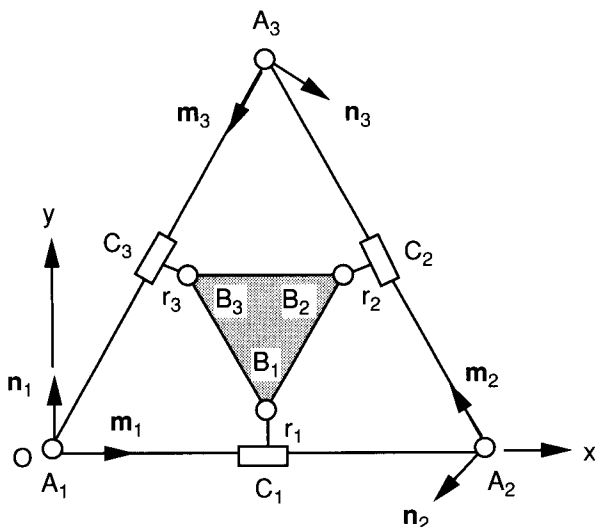


FIGURE 3.18. Equivalent 3RS spatial mechanism.

the base plane:

$$\mathbf{b}_i = \mathbf{c}_i + r_i(\mathbf{n}_i c\phi_i + \mathbf{k} s\phi_i) \quad \text{for } i = 1, 2, 3 \quad (3.135)$$

where $\mathbf{b}_i$ and $\mathbf{c}_i$ denote the position vectors of B_i and C_i , respectively; $\mathbf{n}_i$ denotes a unit vector that lies on the base plane and is perpendicular to line $A_i A_{i+1}$ as shown in Fig. 3.18; and $\mathbf{k}$ is a unit vector pointing in the positive z -axis.

Finally, we formulate the equations associated with the loop closure of the mechanism. Since the distance between points B_i and B_{i+1} is constant, we have

$$[\mathbf{b}_i - \mathbf{b}_{i+1}]^T [\mathbf{b}_i - \mathbf{b}_{i+1}] = h_i^2 \quad \text{for } i = 1, 2, 3 \quad (\text{modulo } 3), \quad (3.136)$$

where $h_i = |B_i B_{i+1}|$. Substituting Eq. (3.135) into (3.136) yields

$$e_{1i} c\phi_i c\phi_{i+1} + e_{2i} s\phi_i s\phi_{i+1} + e_{3i} c\phi_i + e_{4i} c\phi_{i+1} + e_{5i} = 0 \quad (3.137)$$

for $i = 1, 2, 3$ (modulo 3), where

$$e_{1i} = r_i r_{i+1} \mathbf{n}_i^T \mathbf{n}_{i+1},$$

$$e_{2i} = r_i r_{i+1},$$

$$e_{3i} = r_i \mathbf{n}_i^T [\mathbf{c}_{i+1} - \mathbf{c}_i],$$

$$e_{4i} = -r_{i+1} \mathbf{n}_{i+1}^T [\mathbf{c}_{i+1} - \mathbf{c}_i],$$

$$e_{5i} = \frac{h_i^2 - r_i^2 - r_{i+1}^2 - (\mathbf{c}_{i+1} - \mathbf{c}_i)^2}{2}.$$

Equation (3.137) takes the same form as Eq. (3.87). Following the same procedure, Eq. (3.137) can be reduced to an eighth-degree polynomial in the square of one unknown. Therefore, there are at most 16 solutions for the direct kinematics problem. Again, these 16 solutions form eight pairs of manipulator postures, one being the mirror image of another about a plane defined by A_1 , A_2 , and A_3 .

3.11 SUMMARY

Parallel manipulators have been introduced and classified into planar, spherical, and spatial manipulators in accordance with their motion characteristics and their connectivity listings. The Denavit–Hartenberg method for the kinematic analysis of closed-loop mechanisms was reviewed. However, it was judged that the geometric method is more appropriate for the kinematic analysis of most parallel manipulators. The direct and inverse kinematics of several well-known parallel manipulators were analyzed.

The direct kinematics of a 3-dof planar manipulator and a 3-dof orientation platform were both reduced to an eighth-degree polynomial. A spatial parallel manipulators with only translational degrees of freedom was shown to possess two solutions for both the direct and inverse kinematics problems. The direct kinematics of a spatial 3-dof manipulator with coupled position and orientation characteristics was shown to have at most eight pairs of possible poses. Furthermore, the kinematics of a general Stewart–Gough platform and a nearly general Stewart–Gough platform were studied. Finally, a 3–3 Stewart–Gough platform was shown to possess at most eight pairs of possible manipulator postures for its direct kinematics problem. In general, the solutions to the direct kinematics of a parallel manipulator form several pairs of postures, one being the mirror image of the other about some plane of symmetry.

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EXERCISES

1. Sketch as many feasible spatial 3-dof parallel manipulators with a connectivity listing of (5,5,5) as possible. Using the limb configurations shown in Fig. 3.4, identify those mechanisms with identical limb structure.
2. Using the limb configurations shown in Fig. 3.4, sketch as many feasible spatial 4-dof parallel manipulators with a connectivity listing of (6,6,5,5) as possible. Identify those mechanisms with linear input actuators.
3. Referring to the planar five-bar linkage shown in Fig. 1.7, find the end-effector position $\mathbf{q}$ as a function of the two input joint angles, θ_1 and θ_2 .
4. Figure 3.19 shows the schematic diagram of a planar 2-dof five-bar manipulator that is constructed with one prismatic and four revolute joints. Find the end-effector position $\mathbf{q}$ as a function of the two input joint variables, d_1 and θ_2 .
5. Referring to the planar 2-dof pantograph mechanism shown in Fig. 1.8, let 1 and 2 be the input links and 6 be the output link. Also let $BC \parallel DE$, $BE \parallel CD$, and points A , B , and Q lie on a straight line. Find the end-effector position $\mathbf{q}$ as a function of the linear displacements, x_1 and y_2 , of the two input links.

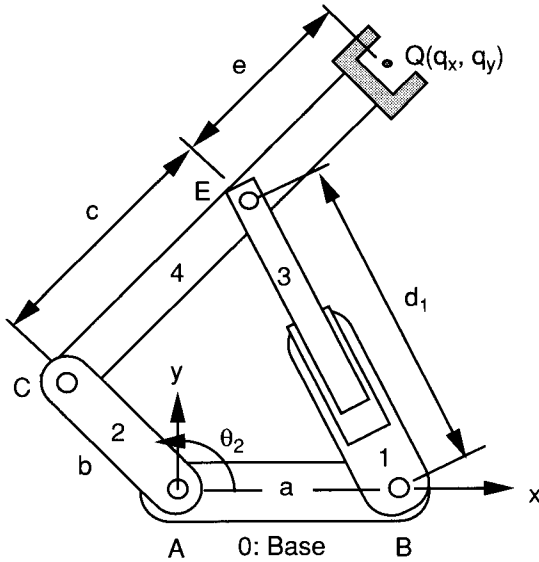


FIGURE 3.19. Five-bar, 4R1P manipulator.

6. Consider the planar 3-dof, 3PRP manipulator shown in Fig. 3.3. Let the three ground-connected prismatic joints be the actuated joints. Assuming that $AB = BC = CA = DE = EF = FD = a$, find the moving platform location in terms of the input displacements.
7. For the spherical 3RRR manipulator shown in Fig. 1.21, let the three ground-connected joints be the input variables. Solve the moving platform orientation as a function of input joint angles. Discuss the number of possible configurations corresponding to a given set of input joint angles.
8. Figure 3.20 shows the schematic diagram of a spatial 3UPU platform manipulator. Three identical limbs connect the moving platform to the fixed base by universal joints at points B_i and A_i , $i = 1, 2$, and 3, respectively. Each limb consists of an upper member and a lower member that are connected by a prismatic joint. The three base-connected axes of the universal joints are coplanar. Similarly, the three moving-platform connected axes of the universal joints are also coplanar. The second axes of the universal joints that are directly attached to the upper and lower members of each limb are parallel to each other and are both perpendicular to the axis of the prismatic joint. It can be shown that this manipulator possesses three translational degrees of freedom. Considering the three prismatic joints as the input joints, solve the direct kinematics problem.

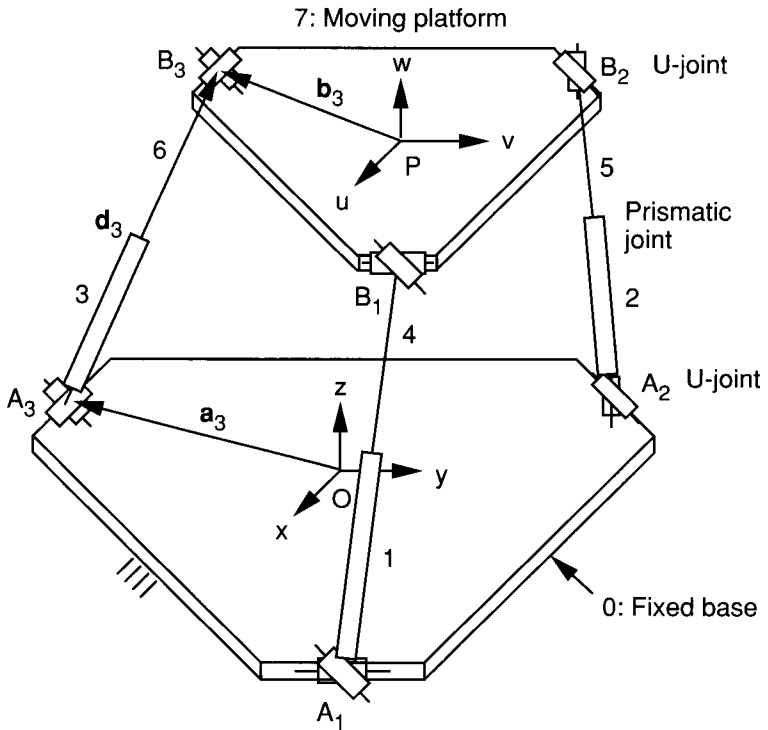


FIGURE 3.20. A 3-dof, 3UPU parallel manipulator.

Also discuss the number of possible configurations corresponding to a given set of input joint variables.

9. Figure 3.21 shows a schematic diagram of the DELTA robot without the gripper (Pierrot et al., 1990, 1991). This robot is made up of a fixed base, a moving platform, and three major limbs. Each major limb connects the moving platform to the fixed platform by an upper arm and a lower arm. The upper arm is made up of a spatial parallelogram (connected by ball joints at the ends of each rod). The moving platform possesses three translational degrees of freedom. Links 1, 2, and 3 are the input links. Three rotary actuators can be mounted on the fixed base to drive the manipulator. Derive the moving platform position $\mathbf{p}$ in terms of the input joint angles.
10. The original flight simulator designed by Stewart (1965) consists of three major limbs, as shown in Fig. 3.22. Each limb consists of two hydraulic jacks. The first jack has its piston connected to the moving platform by a spherical joint, and the cylinder connected to the base by two revolute

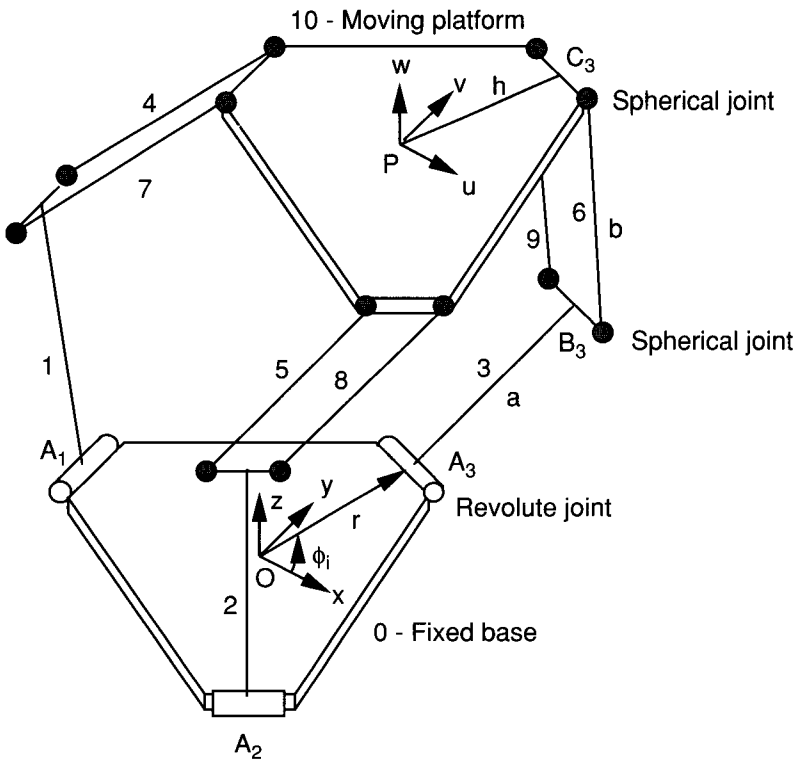


FIGURE 3.21. DELTA platform.

joints. The second jack has its piston connected to the cylinder of the first jack, and the cylinder connected to the base by two revolute joints. The two base-connected revolute joints share one common axis, while the remaining revolute joint axes are parallel to one another and are perpendicular to the base-connected revolute joints. The hydraulic jacks serve as the inputs of the manipulator. Derive the location of the moving platform in terms of the linear displacements of the hydraulic jacks.

11. Figure 3.23 shows a 6-dof parallel manipulator developed by Tsai and Tahmasebi (1993). This manipulator consists of three inextensible limbs. One end of each limb is connected to the moving platform by a revolute joint, while the other end is connected to a 2-dof planar motor by a spherical joint. The three moving-platform-connected revolute joints are coplanar. Derive the moving platform location in terms of the linear displacements of the planar motors.

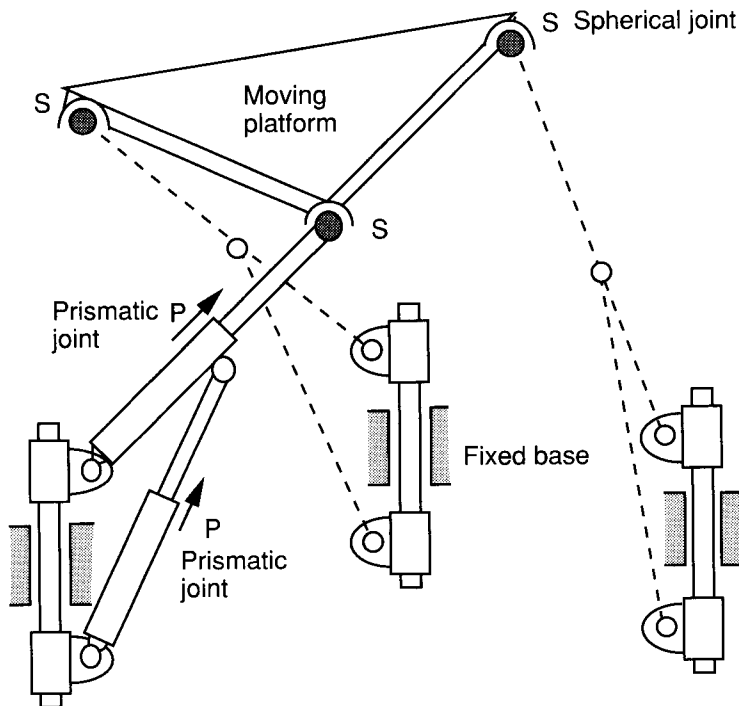


FIGURE 3.22. A 6-dof Stewart platform.

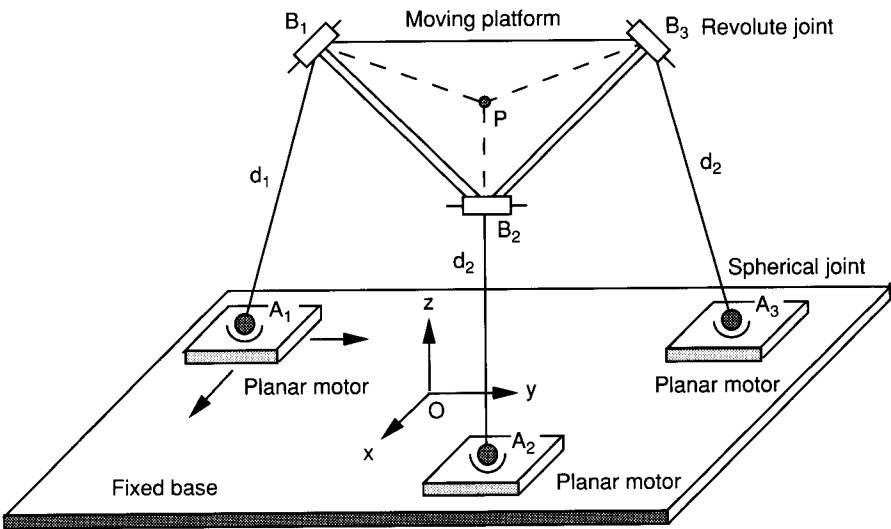


FIGURE 3.23. Spatial 6-dof, 3PPSR platform.